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# DataCollectDefTopic

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## About DataCollect Setup

### About DataCollect Setup

You use DataCollect Setup to specify what data to collect during test program execution and how to display it in a report window or save it in a file.

This overview includes the following topics:

- DataCollect Setup Basics
- Saving and Reusing DataCollect Settings

### Additional DataCollect topics include:

- Setups Tab
- Datalog Setups
- Summary Report, Lot Definition, and Wafer Information Tabs
- Datalog Output Options
- OutputWin Window

|   |                                              |
|---|----------------------------------------------|
| • | <a href="#">Production Control Interface</a> |
|---|----------------------------------------------|

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## About DataCollect Setup : DataCollect Setup Basics

### DataCollect Setup Basics

You use DataCollect Setup to specify what data to collect during test program execution and how to display it in a report window or save it in a file.

### Invoking DataCollect Setup

On the IG-XL Main tab, in the Data Collect group, click DataCollect Setup:



### Data Collect Group

The DataCollect Setup window appears. For an example of the window, see Overview of Using the Setups Tab.

A production interface can also include a button for bringing up DataCollect Setup in production mode. IG-XL Production Controls has a DataCollection control that brings up DataCollect Setup. See DataCollection Control in the Production Controls section.

### Enabling and Disabling Datalog

To enable or disable datalog, do either of the following in DataTool:

- |                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>On the IG-XL Main tab, in the Data Collect group, click Datalog On/Off.</li></ul>                      |
| <ul style="list-style-type: none"><li>In the DataCollect Setup window in the Setups tab, click the Enable data collection control.</li></ul> |

The button toggles between enabled and disabled.

You can enable datalogging for a loaded test program at any time, whether you have set up datalogging or not. If you enable datalogging without altering Datalog Setup, you get a default datalog.

Saved DataCollect Settings

When you exit from an IG-XL workbook, any DataCollect settings you have specified are saved to a file. The next time you open the workbook, the saved settings are retrieved and used for the new session. See [Saving and Reusing DataCollect Settings](#).

Tabs

The DataCollect Setup window includes tabs that let you set up categories of information:

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Setups Tab            | <p>Set up data collection, for datalog or multiple data streams, by selecting standard parametric and functional setups or by creating and selecting customized setups. Setups specify format (what information to report for each test) and criteria (what tests this setup applies to). See <a href="#">Datalog Setups</a>. For functional tests, a customized setup can include HRAM Setup.</p> <p>Control in this tab also allow you to:</p> <ul style="list-style-type: none"><li>• <a href="#">Specify output destinations</a></li><li>• <a href="#">Specify sample rate</a></li><li>• <a href="#">Turn datalogging on or off</a></li></ul> |
| Summary Report Tab    | <p>Set up summary reports, by specifying output destinations and the tests to be included.</p> <p>Generate a partial summary or a final summary.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Lot Definition Tab    | <p>Specify lot header information, plus options relating to disk space, error messages, and the buffer.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Wafer Information Tab | <p>Specify wafer configuration information to be output to an STDF file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

OK, Cancel Buttons

Once you specify the DataCollect setup, click OK. This applies any changes you have made and exits from DataCollect Setup. Do not click OK if you still want to use a different tab.

To quit without applying changes, click Cancel.

Programmatic Interface

VBT provides an object, TheExec.Datalog, that gives programmatic control over datalogging features available from the DataCollect Setup GUI, as well as many additional features. See [Datalog](#)

in the VBT Syntax Reference.

In addition, the DataCollection Production Control has a programmatic interface that corresponds to many of the GUI features. See [Production Control Interface](#).

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# dldatacollect.1.3

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## About DataCollect Setup : Saving and Reusing DataCollect Settings

### Saving and Reusing DataCollect Settings

#### Settings File

When you exit from an IG-XL program, any DataCollect settings you have specified are saved for later reuse with the program.

Note that this feature is available only when the program workbook is open, that is, when you are in Engineering or Offline mode. In Production mode, when the workbook is closed, default DataCollect settings are used.

The settings are saved to a file with the extension .dlex ("datalog extension"). The full filename is: <testprogram-name>\_DL\_Setting.dlex

The file is created in the same folder as the test program.

The next time the test program is opened, DataCollect retrieves the settings stored in this file. To use different settings (for example, to go back to the default settings), use DataCollect to change the settings. These new settings are saved when you exit from the program.

Note the following:

- If you have one workbook open and then open a different workbook, the settings for the first workbook are saved to its file and the settings for the new workbook are retrieved.
- If you use Save As to save a workbook to a new name, the settings are saved to a file whose name is based on the new name, and these settings are retrieved for the new workbook.

### User-Specified Settings

The following settings are saved to the .dlex file:

From the Setups tab (see Overview of Using the Setups Tab):

- [Whether](#) Enable data collection is checked.
- [Whether](#) Customize is checked. That is, whether to use standard datalog setups or customized setups that have been saved in a setup file.
- [For standard datalog setups, whether the](#) Short or the Expanded button is selected and whether Fails Only is checked.
- [If using customized setups, the file that was specified in the](#) Setup File box. The setups saved in this setup file (including parametric setups, functional setups, and the HRAM setup for functional tests) are also retained.
- [Settings for the output destinations for datalogging.](#)
- [Settings of the display options \(specified using the Display Options window, callable from the Setups tab\).](#)

From the Summary Report tab (see [Summary Report Tab](#)):

- [Settings for the output destinations for the summary report.](#)
- [Settings for the test criteria and the output width for the summary report.](#)
- [For](#) Total Summary buttons, whether (By Test) or (By Site) is selected.

From the Lot Definition tab (see [Lot Definition Tab](#)):

- [Options](#) settings.

## Multiple Data Streams

The following settings are saved in the .dlex file to support multiple data streams:

- [Number of Streams](#)
- [Stream Name](#)

Multiple data stream setup information is added at the beginning of the file.

In a .dlex file, multiple data stream setup values are stored in the following sequence:

- Header section of a .dlex file contains the following information:

- File Version Number: All .dlex files have "3.0" as version number in the header section of the file.
- Number of streams: Both datalog and multiple data stream.
- User Name
- Version Number: This value changes every time the structure SSetupDatalogInfo is modified.
- Datalog On: Whether the Enable data collection checkbox is turned on/off.
- Datalog/Data stream: Whether this radio button is selected or deselected

- Stream based settings are stored separately in the following sequence:

- Stream Name: Stores both datalog stream name and multiple data stream name.
- Standard datalog setups
- If using customized setups, the file that was specified in the Setup File box
- Settings for the output destinations for datalogging
- Settings of the display options (specified using the Display Options window)
- Settings for the output destinations for the summary report
- Settings for the test criteria and the output width for the summary report

- [For](#) Total Summary buttons, whether By Test or By Site is selected
- [The](#) Options settings

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## Datalog Output Options

### Datalog Output Options

Choose [output options on the](#) Setups and Summary Report tabs to direct each kind of report to a different output destination.

This section describes the output options. The following topics are available:

- [Output Options](#)
- [STDF Output of DataCollect](#)
- [Custom Output](#)

For more information, see [OutputWin Window](#).

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## Datalog Output Options : Output Options

### Output Options

Choose output options on the [Setups](#) and [Summary Report](#) tabs.  
You can choose more than one output destination on either tab. For example, you can direct datalog output to the window and to a file.  
The following topics are included:

- |   |                                                            |
|---|------------------------------------------------------------|
| • | <a href="#">Output Options</a>                             |
| • | <a href="#">Default Filenames</a>                          |
| • | <a href="#">Report Generation for Window and Text File</a> |
| • | <a href="#">STDF File Destination</a>                      |
| • | <a href="#">When Files Are Closed</a>                      |
| • | <a href="#">Changing Names of Output Files</a>             |
| • | <a href="#">Notes on Window and Text Output</a>            |
| • | <a href="#">Customizing Window and Text Output</a>         |
| • | <a href="#">Postprocessing Datalog Output Files</a>        |

- [DatalogServer Running Out of Memory](#)

For more information, see:

- [Setups Tab](#)

- [Summary Report Tab](#)

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Datalog Output Options : Output Options : Output Options

## Output Options

Output options on the Setups and Summary Report tabs are as follows:

| Output Options          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Window                  | Sends the output to the OutputWin window. See OutputWin Window and Notes on Window and Text Output.                                                                                                                                                                                                                                                                                                                                                                                           |
| Text File,<br>STDF File | <p>Sends the report to an ASCII text file or an STDF file.</p> <p>Enter (or browse to) the pathname of the file to which the output is to be directed. You can enter an absolute or relative pathname; a relative pathname is saved as a relative pathname. When you click OK or Apply, the pathname is validated. An error is generated if the pathname is invalid.</p> <p>See STDF Output of DataCollect and Notes on Window and Text Output.</p> <p>See the note on Default Filenames.</p> |
| Custom Filter           | <p>(Datalog and data stream only - not available for summary) Uses customer-written functions to write the reports. You write these functions as a C++ DLL. See Custom Output.</p> <p>Enter the name of the DLL, followed by the pathname of the file to which the custom output is to be directed.</p>                                                                                                                                                                                       |

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Datalog Output Options : Output Options : Default Filenames

Default Filenames

If you check the option for Text File or STDF File but leave the filename blank, a default filename is used.

For the Datalog stream report, the default filename is:

<workbookname.xls>\_DatalogStream\_ddmmyyyy\_hhmmss.txt or .std

For summary reports, the default filename is:

<workbookname.xls>\_sum\_ddmmyyyy\_hhmmss.txt or .std

This default can cause a problem when the job is executed repeatedly in a loop. If the job execution takes less than a second, then more than one summary report can be written in a second, and the subsequent files overwrite the earlier ones. To avoid this problem, if more than one summary is written in a second, the first summary file has the default format shown above, and subsequent files written during the second have this format:

<workbookname.xls>\_sum\_ddmmyyyy\_hhmmss\_<file number>.txt or .std

The file number starts with 1 and is incremented for each file written during the second. This guarantees unique filenames and avoids overwriting.

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Datalog Output Options : [Output Options](#) : Report Generation for Window and Text File

## Report Generation for Window and Text File

Datalog and summary reports are generated differently, depending on whether the destination is a window or a file.

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Window    | If both the Setups and Summary Report tabs choose Window as the output destination, reports are sent to the same window. A datalog report is written as testing occurs. A summary report is appended when you click Get Partial Summary on the Summary Report tab. A final summary report is appended when you click Get Final Summary/End Lot on the Summary Report tab.                                                            |
| Text File | Records are sent to ASCII text files as specified on the tabs. Datalog records are sent to the text file specified on the Setups tab, and summary records are sent to the text file specified on the Summary Report tab.<br>You cannot send datalog and summary records to the same text file. If you specify the same Text File name on the Setups and Summary Report tabs, the summary text file overwrites the datalog text file. |

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Datalog Output Options : [Output Options](#) : STDF File Destination

STDF File Destination

To create a single STDF file that contains both datalog records and final summary records, specify an STDF File destination on the Setups tab. At end lot, the datalog and final summary records are written to the STDF file you specify.

For a single STDF file, do not check STDF File on the Summary Report tab. In particular, do not specify the same filename on the Setups and Summary Report tabs, on the assumption that this will write datalog and summary records to the same file. Instead, if both tabs specify the same file, the summary STDF file overwrites the datalog STDF file.

To send datalog and summary records to different STDF files, specify different files on the Setups and Summary Report tabs.

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## Datalog Output Options : Output Options : When Files Are Closed

### When Files Are Closed

Output files are closed only on these events:

- |   |                                                                                 |
|---|---------------------------------------------------------------------------------|
| • | EndLot                                                                          |
| • | IG-XL shutdown, if no EndLot has been called                                    |
| • | the output file name is changed and ApplySetup is called                        |
| • | the datalog stream or output is removed from the setup and ApplySetup is called |

When the output files are closed, a Summary event is sent from the datalog service to these data consumers.

Not that a call to TheExec.Datalog.ApplySetup (or the OK button on DataCollect Setup) does not cause the output files to be closed. This means that, if datalogging is on and is then turned off and ApplySetup is called, the output files are not closed. If datalogging is turned on again, new datalog output is appended to the files, as long as the output filenames have not been changed.

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Datalog Output Options : [Output Options](#) : Changing Names of Output Files

Changing Names of Output Files

End Lot and validation both affect the filenames that are used for STDF and text output. Outside the context of End Lot or validation, do not change the STDF or text filename in the middle of a lot run. If you need to do so, you must perform an End Lot to make sure that the data for the current lot is closed out properly. Otherwise the output file may be corrupted. End Lot closes the files and clears out the filenames.

For more information, see [Final Summary/End Lot and Validation](#).

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## Datalog Output Options : Output Options : Notes on Window and Text Output

### Notes on Window and Text Output

This section contains notes on the output that is written to the window and the text file.

#### "G" (Glitch) Notation

A "G" in the output for a functional datalog stands for "glitch." It is used whenever the comparator state is unknown. This may be because the comparator is disabled or because the pin was not at the programmed value for all of the compare window.

#### "A" (Alarm) Notation

An "(A)" in the datalog indicates that the test failed because of an alarm. The alarming test is assigned to the fail bin. See Alarms in the Programming Concepts section.

#### Time Zones

Time zones are not logged. Local time is always assumed. If you change the system's time zone on the fly, you must reload the program to display the correct local time.

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Datalog Output Options : [Output Options](#) : Customizing Window and Text Output

Customizing Window and Text Output

You can use VBT to customize the ASCII output to the window and the text file. Specifically, you can omit columns from the output, and you can specify a nondefault column width. For VBT language, see [TheExec.Datalog.Setup.Shared.Ascii](#) in the VBT Syntax Reference. See also [Notes on Customizing Columns](#).

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## Datalog Output Options : Output Options : Postprocessing Datalog Output Files

### Postprocessing Datalog Output Files

Datalog processing occurs asynchronously, in parallel with IG-XL. This means that the datalog process may still be executing after the End Lot command has returned. For example:

```
TheExec.Datalog.Setup.LotSetup.EndLot = True
```

```
TheExec.Datalog.ApplySetup
```

' Now try processing a resulting output file

After the ApplySetup command returns, the datalog system may still be performing any final processing necessary for the STDF file. "Backgrounding" the datalog processing allows for faster IG-XL performance, but custom post-processing software must take an additional step before attempting to open a datalog output file.

Best programming practice dictates that whenever a process uses a resource that may be shared by a second process or thread, the object's state should always be verified before attempting to access it. To accomplish this in the VBT environment, you can call the following function by passing in the file's path and looping until it returns False.

```
Private Function IsFileAlreadyOpened(fileName As String) As Boolean
```

```
    On Error GoTo Errhandler
```

```
    Open fileName For Input Lock Read Write As #1
```

```
    Close #1
```

```
    IsFileAlreadyOpened = False
```

```
    Exit Function
```

```
Errhandler:
```

```
    IsFileAlreadyOpened = True
```

```
    Exit Function
```

```
End Function
```

Once the function returns "False," you can access the output file for post-processing.

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## Datalog Output Options : Output Options : DatalogServer Running Out of Memory

### DatalogServer Running Out of Memory

The DatalogServer process and the IG-XL process execute in separate threads. If the DatalogServer process runs out of memory, the IG-XL process stops. An error message is returned saying that you must restart IG-XL, and that restarting the IG-XL process also restarts the DatalogServer process. Stopping the IG-XL process when the DatalogServer process runs out of memory ensures that IG-XL does not continue testing after datalog is interrupted.

One reason the DataServer process can run out of memory is the size of data packets that it needs to process. Because some data packets are very large, IG-XL can write these large data packets faster than the DatalogServer can process them, causing the DatalogServer to run out of memory. To prevent this problem, use the VBT statement `TheExec.Datalog.DatalogMaxQueueSize`, which lets you set the maximum size of the data packet queue for both the IG-XL and the DatalogServer process. If either process reaches the maximum queue size, IG-XL pauses until the DataServer has finished processing its queue. Note that setting `DatalogMaxQueueSize` can cause IG-XL to pause during testing, thereby affecting total test time.

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## Datalog Output Options : STDF Output of DataCollect

### STDF Output of DataCollect

DataCollect can write its output as a binary file in Standard Test Data Format (STDF). See [Output Options](#). In addition, IG-XL includes utilities for reading and analyzing STDF files and for converting between a binary STDF file and the ASCII Test Data Format (ATDF).

This section includes these topics:

- [How IG-XL Writes STDF Files](#)
- [Writing Datalog and Summary Data to an STDF File](#)
- [STDF Utilities](#)

On the STDF and ATDF specifications, see:

- [About STDF](#)
- [About ATDF](#)

### How IG-XL Writes STDF Files

IG-XL does not implement all STDF specifications, and it implements some of specifications in ways that differ from other companies and even from other Teradyne products. For more information, see [IG-XL Support for STDF Record Types](#).

### Writing Datalog and Summary Data to an STDF File

Writing datalog data to an STDF file automatically writes summary data to the STDF file. You can, however, write summary data to an STDF file without datalog data. See [Output Options](#).

STDF Utilities

IG-XL includes the following utilities for manipulating STDF files generated by DataCollect. Invoke these utilities from an MS-DOS command line:

|                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ReadStdF [-f]<br>[filename]                                                                | <p>Displays the contents of an STDF file. If you do not specify a filename, input is from standard input.</p> <p>By default, ReadStdF displays the entire contents of each STDF record. If you include the -f switch, it displays only one or two lines of the most important information in each record. It also uses line indentation to indicate record groupings.</p>                                                                        |
| WriteStdF                                                                                  | <p>Creates a sample STDF file called tfile.std.</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| AnalyzeStdF<br>[filename]                                                                  | <p>Displays a brief summary of the contents and size of an STDF file, categorized by record type. If you do not specify a filename, input is from standard input.</p> <p>It displays the size (in bytes) of the entire STDF file. Then, for each record type, it displays the record count, the minimum, average, and maximum record length (in bytes), and the total length of all records of that type.</p>                                    |
| stdfatdf [input-filename] [output-filename]<br>atdfstdf [input-filename] [output-filename] | <p>Convert between STDF and ASCII Test Data Format (ATDF), an ASCII version of STDF:</p> <ul style="list-style-type: none"><li>• stdfatdf converts from STDF to ATDF.</li><li>• atdfstdf converts from ATDF to STDF.</li></ul> <p>If you specify only one filename, it is taken as the input filename, and output is written to standard output. If no filename is specified, input is from standard input and output is to standard output.</p> |



# dloutput.5.14

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## Datalog Output Options : Custom Output

### Custom Output

For custom output, the Flow Controller produces a data stream that is based on STDF record types. User-written functions must check for the type of the current record and then provide code to process the record type. Teradyne-supplied include files define structures for each data type and classes and identifiers for handling the data types.

For more information, see STDF Output of DataCollect.

### Sample Custom Datalog

A sample custom datalog DLL is included in IG-XL as custom.cpp in the folder \src\CustomDatalog. Base your code on this sample DLL.

### Requirements

For custom output, you must write a C++ DLL. The DLL must have these include statements:

- MFC files Afx.h and afxtempl.h
- The Teradyne-supplied file DataType.h located in the folder \IG-XL\  
<version>\src\CustomDatalog\resources.
- The Project Setting of the custom DLL must also include the Teradyne-supplied file DCUtil.lib, located in the folder \IG-XL\  
<version>\lib.

The DLL must contain these functions, which the Flow Controller calls to process the output:

|                  |                                                             |
|------------------|-------------------------------------------------------------|
| CreateCustomFile | Creates the file specified as the custom output file.       |
| WriteCustomData  | Writes data from the data stream to the custom output file. |

|                   |                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FlushCustomBuffer | Writes data from the buffer to disk. This function is called if you select the option Flush buffer during execution. See <a href="#">Lot Definition Tab</a> . |
| CloseCustomFile   | Closes the specified file.                                                                                                                                    |

WriteCustomData Function

This function processes the data stream. As the sample code shows, it is structured as a case statement. Each case checks for the record type of the current record in the data stream, using identifiers defined in the `DataType.h` file. Each case should process that record type by writing some or all of its fields to the output file.

In addition to defining record type identifiers, the `DataType.h` file also defines a set of classes. One of these classes, `CBaseRecord`, has in its protected section an element for testing the current record type.

Each STDF record type also has a separate class. In these classes, a protected element contains a pointer to a C structure for that record type. These structures are defined in the Teradyne-supplied file `stdf.h`, which `DataType.h` includes. (The `stdf.h` file describes each field of each STDF record type and is a useful reference.) The current record is read into the structure for that record type. The case statement you write to process the record can read data from the structure and then write it in the format you want.

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# dloutputwin.6.01

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## OutputWin Window

### OutputWin Window

When DataCollect Setup selects a window as an output option, the OutputWin window is used to display the datalog or summary report. The OutputWin window is described here.

The following topics are available:

- [Using OutputWin](#)
- [OutputWin Menu Commands and Buttons](#)

|                                                                                     |                                                                                     |
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# dloutputwin.6.02



## OutputWin Window : Using OutputWin

### Using OutputWin

The OutputWin window displays a datalog or summary report showing the results of the test program execution. Controls on the window can stop and resume the display and print the report or save it to a file.

The OutputWin window is different from the Test Program Output window, which displays a small set of output, as specified by the Run Options window. See the Test Program Output Window. This topic describes OutputWin window operations. The contents of the report displayed in the window depend on options you select in the DataCollect Setup dialog boxes.

### Invoking the Window

To invoke an OutputWin window, enter the following information in the DataCollect Setup dialog boxes:

- [Check](#) Window in the Setup Type frame of the Setup tab.
- [Check](#) Window in the Output frame of the Summary Report tab.
- [Enable datalogging](#) (using [Enable data collection](#) on the Setups tab or the Datalog On/Off control in the Data Collect group of the IG-XL Main tab) or request a summary report (using buttons on the Summary Report tab).

When you run the program, an OutputWin window is created, if necessary. If a window already exists, the output of the new test program execution is appended to the existing report.

### Stopping Data Collection to Enable Buttons

Use the following controls to stop and resume data collection:



While output is being written to the window, the Stop button is enabled and most of the others are disabled. (About and Help are always enabled.)

This means that, while data is being collected, you cannot use the other buttons to find fails or to save or print the report. To enable these other buttons, stop data collection by clicking Stop.



When you click Stop, the Go button is enabled. The other buttons are enabled as well, so you can use these functions. Once you click Go again, data collection resumes, and all buttons except Stop are disabled again.

Note these additional points:

- In addition to using the Stop and Go buttons, you can also use the Stop and Go commands on the Collect menu.
- When a button is disabled, its corresponding menu command is also disabled. Menu commands without corresponding buttons are disabled as well.
- Once the test program has stopped running and no more data is being collected, the other buttons become enabled. You need not click Stop.

## Zoom Control

A control at the bottom of the window lets you zoom in and out on the data:

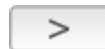
Zoom :  Font size : 10

## Zoom Control

As you move the slider left or right, the Font size: displays the currently selected size.

### ☒ **Note:**

To avoid crashing OutputWin, you must change the default RDC connection options to clear the Printers, Smart cards, Ports, Drives, and Other supported Plug and Play (PnP) devices options in the local devices and resources section of the Local Resources tab. For more information, see [Remote Test System Connections](#).





# dloutputwin.6.03

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## OutputWin Window : OutputWin Menu Commands and Buttons

### OutputWin Menu Commands and Buttons

OutputWin commands and buttons are described here. The following topics are included:

- |   |                                                   |
|---|---------------------------------------------------|
| • | <a href="#">Command Summary</a>                   |
| • | <a href="#">Display Options (Options Command)</a> |
| • | <a href="#">Stop</a>                              |
| • | <a href="#">Go</a>                                |
| • | <a href="#">Find Fail/ALARM</a>                   |
| • | <a href="#">Clear All</a>                         |
| • | <a href="#">Save As</a>                           |
| • | <a href="#">Print</a>                             |
| • | <a href="#">About OutputWin</a>                   |
| • | <a href="#">Help</a>                              |

- [Print Preview](#)
- [Print Setup](#)
- [Exit](#)
- [Toolbar](#)
- [Status Bar](#)
- [Help Topics](#)

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# dloutputwin.6.04

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## OutputWin Window : OutputWin Menu Commands and Buttons : Command Summary

### Command Summary

The following commands are in the OutputWin menus. Most commands have equivalent buttons. Many commands and buttons are disabled when data collection is enabled. To enable them, use the Stop command or button.

#### File Menu

|               |                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Save As       |  <div><p>Saves the report to a file. The command brings up a browser that lets you specify the file.</p><p>See Save As.</p></div> |
| Print         |  <div><p>Prints the report.</p><p>See Print.</p></div>                                                                            |
| Print Preview | <div><p>Displays the report as it would appear printed.</p><p>See Print Preview.</p></div>                                                                                                                           |
| Print Setup   | <div><p>Selects a printer, paper orientation, and other options.</p><p>See Print Setup.</p></div>                                                                                                                    |
| Exit          | <div><p>Exits from OutputWin.</p><p>See Exit.</p></div>                                                                                                                                                              |

#### Edit Menu

**Find FAIL/ALARM**

Finds the next failed test or alarm in the report. At the end of the report, the function loops back to find the first failure or alarm.

See Find Fail/ALARM.

**Clear All**

Clears the window.

See Clear All.

**Collect Menu****Stop**

Stops data collection and enables the buttons that are disabled while data is being collected.

See Stop and Stopping Data Collection to Enable Buttons.

**Go**

Resumes data collection and disables most of the buttons on the toolbar.

See Go.

**Options**

Brings up a dialog box to specify maximum number of lines and whether FAIL appears in red.

See Display Options (Options Command).

**View Menu****Toolbar**

Displays or hides the OutputWin toolbar.

See Toolbar.

**Status Bar**

Displays or hides the status bar at the bottom of the window.

See Status Bar.

**Help Menu****Help Topics**

Displays the online help for OutputWin.

See Help Topics.

**About OutputWin**

Help

Displays information about OutputWin.  
See About OutputWin.



Creates a pointer for context-sensitive help. After clicking this button, move the help pointer to a window object of menu command and click to view help.  
See Help.

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# dloutputwin.6.05

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| <input type="button" value=" &lt;"/> | <input type="button" value=" &gt;"/> |
|--------------------------------------|--------------------------------------|

OutputWin Window : [OutputWin Menu Commands and Buttons](#) : Display Options (Options Command)

## Display Options (Options Command) Options

The Options command on the Collect menu brings up a dialog box that lets you set lines and colors:

|        |                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lines  | The maximum number of lines to be displayed in the report. The limit must be between 1 and 32767. If the report exceeds the specified limit, the leading report lines are overwritten. |
| Colors | Check whether you want the word "FAIL" to be displayed in red. If the box is not checked, "FAIL" is displayed in black.                                                                |

The Options command is enabled when data collection has stopped.

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| <input type="button" value=" &lt;"/> | <input type="button" value=" &gt;"/> |
|--------------------------------------|--------------------------------------|

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## OutputWin Window : OutputWin Menu Commands and Buttons : Stop

### Stop

Stops data collection and enables the buttons that are disabled while data is being collected. While output is being written to the window, only the Stop button is enabled. To enable the other buttons, click Stop. While data collection is enabled, you cannot use the other buttons to find fails or to save or print the report.

When you click Stop, the Go button is enabled. Once you click Go, all buttons except Stop are disabled again.

Stop stops the data collection. It does not pause the test execution. Stopping collection causes some results to be lost. If you use Stop and then Go, data collection is not resumed at the point where it was stopped. Instead, it starts collecting the data that is currently being generated. Any results that would have been collected between the Stop and the Go are, therefore, lost.

Toolbar:



Menu:

Collect

|                 |                 |
|-----------------|-----------------|
| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

# dloutputwin.6.07



## OutputWin Window : OutputWin Menu Commands and Buttons : Go

Go

Resumes data collection and disables most of the buttons on the toolbar.

While output is being written to the window, only the Stop button is enabled. To enable the other buttons, click Stop. While data collection is enabled, you cannot use the other buttons to find fails or to save or print the report.

When you click Stop, the Go button is enabled. Once you click Go, all buttons except Stop are disabled again.

Stop stops data collection. It does not pause the test execution. Stopping collection causes some results to be lost. If you use Stop and then Go, data collection is not resumed at the point where it was stopped. Instead, it starts collecting the data that is currently being generated. Any results that would have been collected between the Stop and the Go are, therefore, lost.

Toolbar:



Menu:

Collect



# dloutputwin.6.08

|                 |                 |
|-----------------|-----------------|
| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

OutputWin Window : [OutputWin Menu Commands and Buttons](#) : Find Fail/ALARM

### Find Fail/ALARM

Finds the next failed test or alarm in the report. At the end of the report, the function loops back to find the first failure or alarm.

This function is available only when data collection has stopped.

Toolbar:



Menu:

Edit

|                 |                 |
|-----------------|-----------------|
| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

# dloutputwin.6.09



OutputWin Window : [OutputWin Menu Commands and Buttons](#) : Clear All

Clear All  
Clears the window.  
[This function is available only when data collection has stopped.](#)

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Toolbar: |  |
| Menu:    | <a href="#">Edit</a>                                                              |



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## OutputWin Window : [OutputWin Menu Commands and Buttons](#) : [Save As](#)

Save As

Saves the report to a file. The function brings up a Browser that lets you specify the file. This function is available only when data collection has stopped.

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Toolbar: |  |
| Menu:    | <a href="#">File</a>                                                              |

|                 |                                          |  |
|-----------------|------------------------------------------|--|
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OutputWin Window : [OutputWin Menu Commands and Buttons](#) : [Print](#)

[Print](#)

Prints the report.

This function is available only when data collection has stopped.

When you select this command, a Print dialog box appears, where you can specify the printer name, the print range, and the number of copies. Related commands are [Print Preview](#) (see [Print Preview](#)) and [Print Setup](#) (see [Print Setup](#)).

While a file is printing, a window appears showing the progress of the printing. To cancel printing, click [Cancel](#).

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| Toolbar: |  |
| Menu:    | <a href="#">File</a>                                                                |

|                                                                                     |                                                                                     |
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| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

[OutputWin Window](#) : [OutputWin Menu Commands and Buttons](#) : [About OutputWin](#)

---

About OutputWin  
Displays information about OutputWin.

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Toolbar: |  |
| Menu:    | <a href="#">Help</a>                                                              |

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OutputWin Window : [OutputWin Menu Commands and Buttons](#) : [Help](#)

Help  
Creates a pointer for context-sensitive help; click the pointer on a window object of menu command.

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Toolbar: |  |
| Menu:    | <a href="#">Help</a>                                                              |

|                 |                                          |  |
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## OutputWin Window : OutputWin Menu Commands and Buttons : Print Preview

### Print Preview

Displays the report as it would appear printed.

When you choose this command, the main window is replaced with a print preview window that displays one or two pages in their printed format. The print preview toolbar has these buttons:

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| Print               | Bring up the print dialog box, to start a print job.          |
| Next Page/Prev Page | Preview the next or previous printed page.                    |
| One Page/Two Page   | Toggle between previewing one or two printed pages at a time. |
| Zoom In/Zoom Out    | Take a closer or larger look at the printed page.             |
| Close               | Return from print preview to the editing window.              |



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OutputWin Window : [OutputWin Menu Commands and Buttons](#) : [Print Setup](#)

Print Setup

Selects a printer, paper orientation, and other options. This command displays a Print Setup dialog box where you specify the printer name, the paper size and source, and the print orientation (portrait or landscape).

This function is available only when data collection is stopped.

|                 |                 |
|-----------------|-----------------|
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OutputWin Window : [OutputWin Menu Commands and Buttons](#) : Exit

Exit

Exits from OutputWin.

You can also exit by clicking the [X](#) in the upper-right corner of the main OutputWin window.



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OutputWin Window : [OutputWin Menu Commands and Buttons](#) : [Toolbar](#)

Toolbar

Displays or hides the OutputWin toolbar. A check mark next to the menu item indicates that the Toolbar is displayed.



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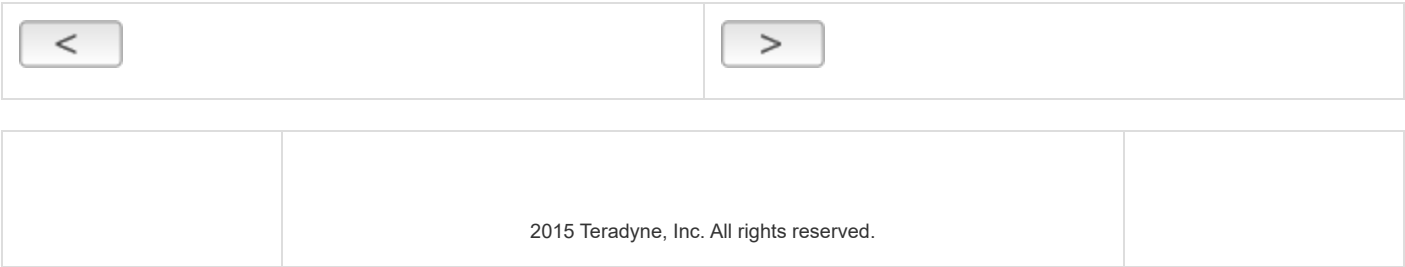
# dloutputwin.6.18



## OutputWin Window : OutputWin Menu Commands and Buttons : Status Bar

### Status Bar

Displays or hides the status bar at the bottom of the window. A check mark next to the menu item indicates that the status bar is displayed.



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|                 |                 |
|-----------------|-----------------|
| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

[OutputWin Window](#) : [OutputWin Menu Commands and Buttons](#) : [Help Topics](#)

[Help Topics](#)  
Displays online help for OutputWin.

|                 |                 |
|-----------------|-----------------|
| <div>&lt;</div> | <div>&gt;</div> |
|-----------------|-----------------|

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## Production Control Interface

### Production Control Interface

VBT provides an object, TheExec.Datalog, that gives programmatic control over datalogging features available from the DataCollect Setup GUI, as well as many additional features. See [Datalog](#) in the VBT Syntax Reference.

In addition, the [Production Control DataCollection Control](#) has a Visual Basic interface that gives programmatic control over features available from the GUI. A programming object is available for each tab in the GUI, plus an interface for sites. This section describes the DataCollection programming interface.

The following topics are available:

- [Programming Interface Overview](#)
- [DatalogSetup Properties](#)
- [SummarySetup Properties](#)
- [LotSetup Properties](#)
- [WaferSetup Properties and Methods](#)
- [SiteSetup Properties](#)

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## Production Control Interface : Programming Interface Overview

### Programming Interface Overview

The set of IG-XL Production Controls includes a [DataCollection](#) control. Adding a [DataCollection](#) control to an operator interface creates a button that, when clicked at run time, brings up the [DataCollect Setup GUI](#). It also gives programmatic access to the data collection setup.

Specifically, an object of type [DataCollection](#) gives access to a set of properties that correspond to the tabs of the [DataCollect Setup GUI](#). (An additional property for sites is available, which does not correspond to a tab.) These properties, in turn, give access to other properties that correspond to specific pieces of the data collection setup, such as whether datalogging has been enabled.

For example, if you have an object named [DataCollection1](#), of type [DataCollection](#), you can use this statement to set the tester mode:

```
DataCollection1.LotSetup.TestMode = TestModeProd
```

Using this interface, you can programmatically set up data collection and output, as well as provide information to be written to an output file.

For more information, see [DataCollection Control](#).

### DataCollection Properties

The properties for an object of type [DataCollection](#) are as follows.

| DataCollection Programming Objects |                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------|
| Property                           | Description                                                                                          |
| <a href="#">DatalogSetup</a>       | Corresponds to the <a href="#">Setups</a> tab. See <a href="#">DatalogSetup Properties</a> .         |
| <a href="#">SummarySetup</a>       | Corresponds to the <a href="#">Summary Report</a> tab. See <a href="#">SummarySetup Properties</a> . |
| <a href="#">LotSetup</a>           | Corresponds to the <a href="#">Lot Definition</a> tab. See <a href="#">LotSetup Properties</a> .     |

|            |                                                                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| WaferSetup | Corresponds to the Wafer Information tab. A set of WaferSetup methods is also available. See <a href="#">WaferSetup Properties and Methods</a> . |
| SiteSetup  | Does not correspond to any tab; provides a means to write site description information. See <a href="#">SiteSetup Properties</a>                 |

Applying the Setup

DataCollect Setup properties do not take effect until you execute the following method of the DataCollection production control:

[DataCollection1.ApplySetup](#)

This method applies the data collection properties that you specified and is equivalent to clicking OK on the interactive GUI. Note that the DataCollection method .EndLot also executes the .ApplySetup method.

For more information, see:

- [DataCollection Control](#)
- [EndLot Method](#)

Correspondences Between Property Types and Tab Items

Most properties correspond to check boxes, input fields, and drop-down boxes on the tabs of the DataCollect Setup dialog boxes:

- [Any property whose type is Boolean](#) corresponds to a check box. True means checked, and False means not checked.
- [Any property whose type is String or numeric](#) corresponds to an input field. Assign a string or number as appropriate.
- [Any property whose type is Enum](#) corresponds to a drop-down list. The enumerated values correspond to the values available on the drop-down list.

A few properties do not correspond to anything on the tabs. These are based on useful STDF fields and can be used to write additional information to an STDF output file.

Compatibility with Previous Releases of DataCollect Setup

In previous releases of IG-XL, some DataCollect Setup controls appear on different tabs. For example, the Enable data collection check box (formerly the Datalog On check box), which is now on the Setups tab, previously appeared on the Lot Setup tab. In this previous release, the

.DatalogOn property belonged to LotSetup; to correspond to the new setup, it should belong to DatalogSetup.

To maintain compatibility with existing programs, the DataCollect Setup programming interface supports such properties in both its previous and its new location. For example, .DatalogOn is a property of both DatalogSetup and LotSetup. In the descriptions, the properties are first listed under their current location. A separate table lists “obsolete” properties, that is, properties that were on the tab corresponding to this object but that are now on a different tab.

Note on Type-Ahead

Because of an implementation constraint, the DataCollect Setup interface does not support type-ahead below the level of the properties listed above. That is, when you type in the name of the DataCollection object, a drop-down list displays the properties shown above. When you then select one of these properties, however, no further drop-down list appears. You must type in the property name.

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# dlprogramming.7.3

|                                                                          |                                                                          |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <input style="width: 50px; height: 20px;" type="button" value=" &lt; "/> | <input style="width: 50px; height: 20px;" type="button" value=" &gt; "/> |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|

## Production Control Interface : DatalogSetup Properties

### DatalogSetup Properties

The DatalogSeup property of a DataCollection object gives access to the datalog-specific settings for data collection.

### Datalog Setup Files

The programming interface does not let you specify the details of datalogging, which are stored in the datalog setup files. Use the DataCollect Setup GUI to create, select, and save setup files.

The programming interface does contain properties to enable use of an existing setup file (SelectSetupFile) and, if enabled, to specify the file (SetupFile). The setups that were previously selected using the GUI are used for the current datalogging session. (When you save a setup file, the information about which setups are currently selected is also saved.)

For more information, see [Datalog Setups](#).

### Applying the Setup

The DatalogSetup properties do not take effect until you execute the ApplySetup method of the TesterControl production control. This method applies the specified data collection properties and is equivalent to clicking OK on the interactive GUI.

For more information, see [ApplySetup Method](#).

### Note on Type-Ahead

Because of an implementation constraint, the DataCollect Setup interface does not support type-ahead below the level of the DatalogSetup property. You must type in the properties listed below.

### Properties

The following table lists the DatalogSetup properties.

All DatalogSetup properties correspond to fields on the Setups tab. Properties relating to output formatting and content correspond to fields on the Display Options window, accessed by the Display Settings button on the Setups tab.

For more information, see:

- [Setups Tab](#)
- [Display Settings](#)

DatalogSetup Properties

| Property         | Type/Description                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------|
| CustomOutput     | Boolean<br>Whether the Custom Filter output box is checked.                                             |
| CustomOutputFile | String<br>If CustomOutput is true, the value entered as the DLL name and output filename.               |
| DatalogOn        | Boolean<br>Whether the Enable data collection box is checked, that is, whether datalogging is enabled.  |
| DisableDtr       | Boolean<br>Whether the Disable Datalog Text Record output box is checked in the Display Options window. |
| HeaderEveryRun   | Boolean<br>Whether the Header Every Time box is checked in the Display Options window.                  |
| OutputWidth      | Long<br>The value entered in the Output Width field in the Display Options window.                      |
| PartResult       | Boolean<br>Whether the Part Result box is checked in the Display Options window.                        |
| SelectSetupFile  | Boolean<br>Whether the Customize box is checked.                                                        |
| SetupFile        | String<br>If SelectSetupFile is true, the value entered as the filename.                                |
| STDFOutput       | Boolean<br>Whether the STDF File box is checked.                                                        |

|                |                                                                                         |
|----------------|-----------------------------------------------------------------------------------------|
| STDFOutputFile | String<br>If STDFOutput is checked, the value entered as the filename.                  |
| Synopsis       | Boolean<br>Whether the Test Synopsis Data box is checked in the Display Options window. |
| TextOutput     | Boolean<br>Whether the Text File box is checked.                                        |
| TextOutputFile | String<br>If TextOutput is true, the value entered as the filename.                     |
| WindowOutput   | Boolean<br>Whether the Window output box is checked.                                    |

For more information, see [Setups Tab](#).

[Obsolete Datalog Setup Tab Properties](#)

The following properties correspond to fields that in previous releases are on the Datalog Setup tab, but that in the current release have been moved to another tab. For compatibility, these properties are supported both under Datalog Setup (DatalogSetup) and under the new tab.

Obsolete DatalogSetup Properties

| Property    | Type/Description                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------|
| SampleLimit | Long<br>The value entered in the Sample Limit field.<br>Moved to LotSetup. See <a href="#">LotSetup Properties</a> . |
| SampleRate  | Long<br>The value entered in the Sample Rate field.<br>Moved to LotSetup. See <a href="#">LotSetup Properties</a> .  |

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## Production Control Interface : SummarySetup Properties

### SummarySetup Properties

The SummarySetup property of a DataCollection object gives access to the summary-specific settings for data collection. All SummarySetup properties correspond to fields on the Summary Report tab.

For more information, see [Summary Report Tab](#).

### Applying the Setup

SummarySetup properties do not take effect until you execute the ApplySetup method of the TesterControl production control. This method applies the specified data collection properties and is equivalent to clicking OK on the interactive GUI.

For more information, see [ApplySetup Method](#).

### Note on Type-Ahead

Because of an implementation constraint, the DataCollect Setup interface does not support type-ahead below the level of the SummarySetup property. You must type in the properties listed below. Properties

The following table lists the SummarySetup properties.

| SummarySetup Properties |                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Property                | Type/Description                                                                                                                                                               |
| CriteriaType            | Enum<br>The value selected from the Test Criteria drop-down list. The enumerated values are: <ul style="list-style-type: none"><li>TL_DCALLTESTS: Include all tests.</li></ul> |

- [TL\\_DCFAILTESTS](#): Include only failed tests.
- [TL\\_DCNOTESTS](#): Do not include any tests.

[CustomOutput](#)

Boolean

No longer supported in the [DataCollect Setup](#) interface.[EndLot](#)

Boolean

Whether a final summary is to be generated.

If true, a final summary is generated and the STDF and text files are finished and closed when the setup is applied.

Note that the effect of this property is different from clicking [Get Final Summary/End Lot](#) on the Summary Report tab. Clicking the button generates the report immediately, whereas the property does not take effect until [.ApplySetup](#) is executed.

[CustomOutputFile](#)

String

No longer supported in the [DataCollect Setup](#) interface.[OutputWidth](#)

Long

The value entered in the [Output Width](#) field.[SiteReport](#)

Boolean

Whether the [Total Summary \(By Site\)](#) radio button is selected.[STDFOutput](#)

Boolean

Whether the [STDF File](#) box is checked.[STDFOutputFile](#)

String

If [STDFOutput](#) is true, the value entered as the filename.[SummaryOn](#)

Boolean

Whether a partial summary report is to be generated.

Note the following:

- The effect of this property is different from clicking [Get Partial Summary](#) on the Summary Report tab. Clicking the button generates the report immediately, whereas the property does not take effect until [.ApplySetup](#) is executed.
- The [GetSummaryNow](#) method of the [DataCollection](#) production control is equivalent to [.SummaryOn](#) followed by

[.ApplySetup.](#)

|                                |                                                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">TextOutput</a>     | <a href="#">Boolean</a><br><a href="#">Whether the</a> Text File box is checked.                                                      |
| <a href="#">TextOutputFile</a> | <a href="#">String</a><br>If <a href="#">TextOutput</a> is <a href="#">true</a> , the <a href="#">value entered as the filename</a> . |
| <a href="#">WindowOutput</a>   | <a href="#">Boolean</a><br><a href="#">Whether the</a> Window output box is checked.                                                  |

For more information, see [Summary Report Tab](#).

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## Production Control Interface : LotSetup Properties

### LotSetup Properties

The LotSetup property of a DataCollection object gives access to the lot-wide settings for data collection.

The two sets of LotSetup properties are:

- Properties that correspond to fields on the Lot Definition tab
- Properties that correspond to fields on the More Information dialog box, which is invoked from the Lot Definition tab

LotSetup properties are given below in separate lists. In the programming interface, they are all properties of the LotSetup object.

Some Lot tab properties and all More Information properties correspond to fields in the STDF Master Information Record (MIR).

For more information, see:

- Lot Definition Tab Properties
- More Information Properties
- Master Information Record (MIR)

### Applying the Setup

The LotSetup properties do not take effect until you execute the ApplySetup method of the TesterControl production control. This method applies the specified data collection properties and

is equivalent to clicking OK on the interactive GUI.

For more information, see [ApplySetup Method](#).

Note on Type-Ahead

Because of an implementation constraint, the DataCollect Setup interface does not support type-ahead below the level of the LotSetup property. You must type in the properties listed below.

Lot Definition Tab Properties

The following table lists the LotSetup properties. These properties correspond to fields on the Lot Definition tab.

| LotSetup Properties for the Lot Definition Tab |                                                      |
|------------------------------------------------|------------------------------------------------------|
| Property                                       | Type/Description                                     |
| DeviceNumber                                   | Long<br>Value entered in the Device Number field.    |
| LotID                                          | String<br>Value entered in the Lot ID field.         |
| Operator                                       | String<br>Value entered in the Operator field.       |
| PartType                                       | String<br>Value entered in the Part Type field.      |
| Retest                                         | Boolean<br>Not currently used.                       |
| SampleLimit                                    | Long<br>The value entered in the Sample Limit field. |
| SampleRate                                     | Long<br>The value entered in the Sample Rate field.  |

|          |                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| TestMode | Enum                                                                                                                                                      |
|          | Value selected for the Test Mode field. The enumerated type for this property has these defined values:                                                   |
|          | <ul style="list-style-type: none"><li>TestModeEng: Engineering mode</li><li>TestModeProd: Production mode</li><li>TestModeNone: No mode defined</li></ul> |

For more information, see [Lot Definition Tab](#).

Obsolete Lot Definition Tab Properties

The following properties correspond to fields that in previous releases are on the Lot Definition tab, but that in the current release have been moved to another tab. For compatibility, these properties are supported both under Lot Definition (LotSetup) and under the new tab.

Obsolete LotSetup Properties

| Property     | Type/Description                                                                                                                                                                                                                                 |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DatalogOn    | Boolean<br>Whether the Datalog box is checked, that is, whether datalogging is enabled.<br>Moved to DatalogSetup. See <a href="#">DatalogSetup Properties</a> .                                                                                  |
| EndLot       | Boolean<br>Whether the End Lot box is checked.<br>If true, a final summary is generated, and the STDF and text files are finished and closed, when the setup is applied.<br>Moved to SummarySetup. See <a href="#">SummarySetup Properties</a> . |
| SummaryOn    | Boolean<br>Whether the Get Summary now box is checked.<br>Moved to SummarySetup. See <a href="#">SummarySetup Properties</a> .                                                                                                                   |
| StdDirectory | String<br>Value entered in the STDF File default directory field.<br>Not used in the current DataCollect Setup interface.                                                                                                                        |

|               |                                                         |
|---------------|---------------------------------------------------------|
| TextDirectory | String                                                  |
|               | Value entered in the Text File default directory field. |
|               | Not used in the current DataCollect Setup interface.    |

More Information Properties

These properties correspond to fields on the More Information dialog box, invoked from the Lot Definition tab. See [More Information Dialog Box](#), which allows additional values to be written to the output STDF file and, optionally, to the window and a text file.

Note that there is an additional property for each “More Information” property. This property, prefixed with a “b,” is a Boolean value that corresponds to the check box next to the field in the More Information dialog box. If the box is checked (that is, the Boolean property is true), the value is written to the window and to the text file, as well as to the STDF file.

The list of properties below does not include these “b” properties.

The More Information dialog box also contains a tab that lets you write information to the Site Description Record (SDR). The programmatic interface supports these SDR fields using the SiteSetup programming object. See [SiteSetup Properties](#).

The properties (excluding “b” properties, as described above) are as follows. Each property is of type String; the corresponding “b” property is of type Boolean. For more information, see [Master Information Record \(MIR\)](#) in the STDF Specification.

LotSetup Properties for the More Dialog Box

| Property   | Description                              |
|------------|------------------------------------------|
| AuxFile    | Auxiliary data file                      |
| BurnTime   | Burn-in time                             |
| Command    | Command mode code                        |
| DateCode   | Date code                                |
| DesignRev  | Device design revision                   |
| EngID      | Engineering lot ID                       |
| ExecType   | Tester executive software type           |
| ExecVer    | Tester executive software version number |
| FacilityID | Test facility ID                         |
| FamilyID   | Product family ID                        |
| FloorID    | Test floor ID                            |

|            |                                   |
|------------|-----------------------------------|
| FlowID     | Test flow ID                      |
| JobRev     | Job revision number               |
| OperFrq    | Operation frequency or step       |
| PkgType    | Package type                      |
| ProcessID  | Fabrication process ID            |
| ProtCode   | Data protection code              |
| RetestCode | Lot retest code                   |
| RomID      | ROM code ID                       |
| SerialNum  | Tester serial number              |
| SetupID    | Test setup ID                     |
| SpecName   | Test specification name           |
| SpecVer    | Test specification version number |
| SubLot     | Sublot ID                         |
| Supervisor | Supervisor name or ID             |
| TestCode   | Test phase or step code           |
| TstTemp    | Test temperature                  |
| UserText   | Generic user text                 |

|                                           |                                          |  |
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## Production Control Interface : WaferSetup Properties and Methods

### WaferSetup Properties and Methods

The `WaferSetup` property of a `DataCollection` object gives access to wafer-specific settings for data collection.

All `WaferSetup` properties correspond to check boxes, input fields, and drop-down lists on the Wafer Information tab. The values correspond to fields in the STDF Wafer Configuration Record (WCR).

In addition to the properties listed here, the `WaferSetup` object also gives access to a set of methods for getting and setting the X and Y coordinates.

For more information, see:

- |   |                                                  |
|---|--------------------------------------------------|
| • | <a href="#">Wafer Information Tab</a>            |
| • | <a href="#">Wafer Configuration Record (WCR)</a> |

### Applying the Setup

The `WaferSetup` properties do not take effect until you execute the `ApplySetup` method of the `TesterControl` production control. This method applies the specified data collection properties and is equivalent to clicking OK on the interactive GUI.

For more information, see [ApplySetup Method](#).

### Note on Type-Ahead

Because of an implementation constraint, the `DataCollect Setup` interface does not support type-ahead below the level of the `WaferSetup` property. You must type in the properties listed below.

### Properties

The following table lists the `WaferSetup` properties.

## WaferSetup Properties

| Property  | Type/Description                                                                                                                                                                                                                                                                 |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diameter  | <p>Single</p> <p>The value entered in the Diameter of Wafer field (units given by Units).</p>                                                                                                                                                                                    |
| DieHeight | <p>Single</p> <p>The value entered in the Height of Die field (units given by Units).</p>                                                                                                                                                                                        |
| DieWidth  | <p>Single</p> <p>The value entered in the Width of Die field (units given by Units).</p>                                                                                                                                                                                         |
| ID        | <p>String</p> <p>The value entered in the Wafer ID field.</p>                                                                                                                                                                                                                    |
| Orient    | <p>Enum</p> <p>The value selected from the drop-down list for Orientation of Wafer Flat. The enumerated values are:</p> <ul style="list-style-type: none"><li>• TL_WCRDOWN</li><li>• TL_WCRUP</li><li>• TL_WCRLEFT</li><li>• TL_WCRRIGHT</li><li>• TL_WCRUNKNOWNORIENT</li></ul> |
| Unit      | <p>Enum</p> <p>The value selected from the drop-down list for Unit. The enumerated values are:</p> <ul style="list-style-type: none"><li>• TL_WCRCENTIMETERS</li><li>• TL_WRINCHES</li><li>• TL_WCRMILLIMETERS</li></ul>                                                         |

- [TL\\_WCRMILS](#)
- [TL\\_WCRUNKNOWNUNITS](#)

**XCenter**

Integer

The value entered in the X field in the Coordinate of Center Die area.

**XPos**

Enum

The value selected from the drop-down list for X in the Direction of Wafer area. The enumerated values are:

- [TL\\_WCRLEFT](#)
- [TL\\_WCRRIGHT](#)
- [TL\\_WCRUNKNOWNORIENT](#)

**YCenter**

Integer

The value entered in the Y field in the Coordinate of Center Die area.

**YPos**

Enum

The value selected from the drop-down list for Y in the Direction of Wafer area. The enumerated values are:

- [TL\\_WCRDOWN](#)
- [TL\\_WCRUP](#)
- [TL\\_WCRUNKNOWNORIENT](#)

For more information, see [Wafer Information Tab](#).

## Methods

The methods for the [WaferSetup](#) object are as follows:

## WaferSetup Methods

| Name                                              | Type     |
|---------------------------------------------------|----------|
| GetXCoord(nSite As Integer) As Integer            | Function |
| GetYCoord(nSite As Integer) As Integer            | Function |
| SetXCoord(nSite As Integer, nNewValue As Integer) | Sub      |
| SetYCoord(nSite As Integer, nNewValue As Integer) | Sub      |

**Get** functions get the coordinate of the device currently being tested on the specified site.

**Set** functions set the coordinate of the device currently being tested on the specified site to the specified new value.

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## Production Control Interface : SiteSetup Properties

### SiteSetup Properties

The SiteSetup property of a DataCollection object gives access to settings that relate to site configuration. Use these properties to write site-related information to an STDF file. The properties correspond to the fields of a Site Description Record (SDR).

The SiteSetup object corresponds to the SDR tab of the More Information dialog box, accessed by the More button on the Lot Definition tab.

For more information, see:

- [Site Description Record \(SDR\)](#)
- [More Information Dialog Box](#)
- [LotSetup Properties](#)

### Type and ID Properties

The SiteSetup properties are paired:

- [Properties ending with Type](#) are the type or model number of the interface or peripheral equipment being used for testing.
- [Properties ending with ID](#) are the IDs or serial numbers of the specific piece of equipment.

### "b" Properties

Note that there is an additional property for each Type and ID property. This property, prefixed with a "b," is a Boolean value that is modelled on the check box next to each field on the More

Information dialog box. If the “b” property is True, the value is written to the window and to the text file, as well as to the STDF file.

The list of properties below does not include these “b” properties.

Applying the Setup

The SiteSetup properties do not take effect until you execute the ApplySetup method of the TesterControl production control. This method applies the specified data collection properties and is equivalent to clicking OK on the interactive GUI.

For more information, see ApplySetup Method.

Note on Type-Ahead

Because of an implementation constraint, the DataCollect Setup interface does not support type-ahead below the level of the SiteSetup property. You must type in the properties listed below.

Properties

The properties for the SiteSetup object (excluding “b” properties, as described above) are as follows.

| SiteSetup Properties |                                  |
|----------------------|----------------------------------|
| Property             | Type/Description                 |
| CablID               | String<br>Interface cable        |
| CablType             | String                           |
| CardID               | String<br>Probe card             |
| CardType             | String                           |
| ContID               | String<br>Handler contactor      |
| ContType             | String                           |
| DibID                | String<br>Device interface board |
| DibType              | String                           |
| ExtrID               | String<br>Extra equipment        |
| ExtrType             | String                           |

|          |                             |
|----------|-----------------------------|
| HandID   | String<br>Handler or prober |
| HandType | String                      |
| LasrID   | String<br>Laser             |
| LasrType | String                      |
| LoadID   | String<br>Load board        |
| LoadType | String                      |

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## Setups Tab

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### Setups Tab

This section describes the Setups tab of DataCollect Setup. The following topics are covered:

- |   |                                                  |
|---|--------------------------------------------------|
| • | <a href="#">Overview of Setups and Streams</a>   |
| • | <a href="#">Overview of Using the Setups Tab</a> |
| • | <a href="#">Working with Data Streams</a>        |
| • | <a href="#">Specifying Setups</a>                |
| • | <a href="#">Specifying Output</a>                |
| • | <a href="#">Specifying Sampling</a>              |
| • | <a href="#">General Controls</a>                 |
| • | <a href="#">Display Settings</a>                 |

For information on other DataCollect tabs, see [Summary Report](#), [Lot Definition](#), and [Wafer Information Tabs](#).

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## Setups Tab : Overview of Setups and Streams

### Overview of Setups and Streams

The Setups tab uses the concepts of datalog setups and data streams. This section defines these terms. The Current Data Streams section of the Setups tab looks like this:

Current Data Streams

HRAM Setup: DefaultHRAMSetup

| Type        | Stream Name   | Output | Parametric Info    | Functional Info      | Sample Rate | Sample Lim |
|-------------|---------------|--------|--------------------|----------------------|-------------|------------|
| Datalog     | DatalogStream | WIN    | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |
| Data stream | MDSFails      | STDF   | C:\MP4\DIgSt.stp   | C:\MP4\DIgSt.stp     | 1           | 0          |
| Data stream | MDSAll        | STDF   | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |

New

Delete

### Current Data Streams

A datalog setup specifies:

- The criteria for selecting tests to be datalogged: for example, all tests, all failing tests, a list of test numbers or names. By default, all tests are datalogged. You can easily specify that you want only failed tests datalogged. You can also create a custom setup to specify which tests you want datalogged.
- The format for presenting datalogged information. DataCollect has the built-in formats Short and Expanded. It also lets you specify a custom format.

The Setups tab has controls for selecting the built-in short or expanded formats or to say that you want to datalog only fails. It also has a window for specifying a custom setup. You can save the custom setups to a file for reuse in later sessions.

A data stream is data collection directed toward a specific output destination. Each user-defined data stream consists of:

- A datalog setup, as described above, specifying the test criteria and the output format. This can be either one of the built-in setups or a custom setup.
- An output destination. This can be a text file or an STDF file. A data stream can have only one output destination. The data stream can also specify a user-created custom filter to process the data being sent to the output destination.
- Sampling information. This is the sample rate and the sample limit.

You can have as many data streams as you like. Data streams let you create multiple output files, each having exactly the information and format you want, without having to perform postprocessing.

In addition to user-created data streams, DataCollect contains a stream named DatalogStream that is the default datalogging specification. It can direct output to the datalog window as well as to an STDF file and text file; whereas a user-defined data stream can specify one only output destination, DatalogStream can specify all three output destinations. You can modify the default DatalogStream specification.

When you execute a program with data collection turned on, all setups currently displayed on the Setups tab are used to control data collection and datalog output.

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## Setups Tab : Overview of Using the Setups Tab

### Overview of Using the Setups Tab

You use the Setups tab to perform the following actions:

- [Enable or disable data collection. See Enable data collection.](#)
- [Specify the action to be taken on validation or save. See Following Validate/Save.](#)
- [Specify a nondefault HRAM setup. See HRAM Setup in DataCollect Setup.](#) This setting applies to all data streams.
- [Create custom-defined data streams, and if desired modify the built-in DatalogStream.](#) This involves specifying the setup, the output destination, and sampling. See [Working with Data Streams](#). These controls take up most of the tab.
- [Control the way datalogged information is formatted and displayed. See Display Settings.](#) These settings apply to all data streams.

In general, controls are divided between those controls that define a specific data stream and those that apply to all data streams. See [Working with Data Streams](#) and [General Controls](#).  
The Setups tab looks like this:

DataCollect Setup - C:\MP4\DlgSt.stp

Setups | Summary Report | Lot Definition | Wafer Information

☒ Enable data collection      Following Validate/Save: Close Datalog/Discard Summary

Current Data Streams      HRAM Setup: DefaultHRAMSetup

| Type        | Stream Name   | Output | Parametric Info    | Functional Info      | Sample Rate | Sample Lim |
|-------------|---------------|--------|--------------------|----------------------|-------------|------------|
| Datalog     | DatalogStream | WIN    | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |
| Data stream | MDSFails      | STDF   | C:\MP4\DlgSt.stp   | C:\MP4\DlgSt.stp     | 1           | 0          |
| Data stream | MDSAll        | STDF   | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |

New      Delete

Setup Type

☐ Datalog      ☒ Data Stream

☐ Window      Log By Test      Name: MDSFails

Setups

Short      Expanded      Fails Only

Parametric: ☐ ☒ ☒      ☒ Customize: Setup File...

Functional: ☐ ☒ ☒      Setup File: C:\MP4\DlgSt.stp      Browse...

Output

☐ Text File      Log By Test      Browse...

☐ OverWrite File

☒ STDF File      Log By Test      Browse...

☐ OverWrite File      ☐ Disable Datalog Text Record

☐ Custom Filter      Browse...

Sample

Sample Rate: 1      Sample Limit: 0

☐ Enable Per-Test Sampling      Select Tests ...

Display Settings...      Apply      Reset

OK      Cancel      Help

## DataCollect Setups Tab

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Setups Tab : Working with Data Streams

Working with Data Streams

The primary function of the Setups tab is to create and manage data streams, including the built-in datalog stream.

The following topics are covered:

- |   |                                                 |
|---|-------------------------------------------------|
| • | Current Data Streams                            |
| • | Creating, Modifying, and Deleting a Data Stream |
| • | Modifying the Built-In DatalogStream            |
| • | Saving the Setup with Apply                     |
| • | Resetting Data Streams                          |

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Setups Tab : Working with Data Streams : Current Data Streams

### Current Data Streams

The Current Data Streams grid looks like this:

Current Data Streams

HRAM Setup: DefaultHRAMSetup

| Type        | Stream Name   | Output | Parametric Info    | Functional Info      | Sample Rate | Sample Lim |
|-------------|---------------|--------|--------------------|----------------------|-------------|------------|
| Datalog     | DatalogStream | WIN    | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |
| Data stream | MDSFails      | STDF   | C:\MP4\DlgSt.stp   | C:\MP4\DlgSt.stp     | 1           | 0          |
| Data stream | MDSAll        | STDF   | ALL TESTS + EXP... | ALL TESTS + EXPANDED | 1           | 0          |

NewDelete

### Current Data Streams Grid

The grid displays data streams currently defined for this workbook, including the built-in DatalogStream. Data stream definitions are stored in a .dlex file in the workbook folder. When you open the Setups tab, the definitions currently in the .dlex file are loaded into the grid.

The grid is read-only, and you cannot directly edit it. Instead, you use controls on the tab to add data streams as well as to edit or delete the currently selected data stream.

A tool tip is provided every time you move over any cell in the grid to show the content of the cell. This is useful when the content of a cell is wider than the display.

The grid in the Current Data Streams frame provides the following information:

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                               | <p>The type of stream:</p> <ul style="list-style-type: none"> <li>• <a href="#">Datalog</a> for the built-in DatalogStream</li> <li>• <a href="#">Data stream</a> for custom data streams</li> </ul>                                                                                                                                                                                                                                                                                         |
| Stream Name                        | <p>The name of the data stream. DatalogStream is the name of built-in stream. Custom data streams have user-assigned names.</p>                                                                                                                                                                                                                                                                                                                                                              |
| Output                             | <p>The output destination specified for the stream, as follows:</p> <ul style="list-style-type: none"> <li>• <a href="#">WIN</a>: Output Window. DatalogStream only; not available for custom data streams.</li> <li>• <a href="#">STDF</a>: STDF file.</li> <li>• <a href="#">TEXT</a>: ASCII text file.</li> <li>• <a href="#">+CUSTOM</a>: A custom filter has been specified in addition to the output destination.</li> </ul> <p>See <a href="#">Specifying Output</a>.</p>             |
| Parametric Info<br>Functional Info | <p>The parametric and functional setup information, as follows:</p> <ul style="list-style-type: none"> <li>• <a href="#">ALL TEST</a> or FAIL TEST: whether the Fails Only check box is checked.</li> <li>• <a href="#">+ SHORT</a> or + EXPANDED: whether the Short or Expanded built-in setup radio button has been selected.</li> <li>• <a href="#">If a custom setup is used, the setup file pathname appears in this cell.</a></li> </ul> <p>See <a href="#">Specifying Setups</a>.</p> |
| Sample Rate<br>Sample Limit        | <p>The sample rate and sample limit. See <a href="#">Specifying Sampling</a></p>                                                                                                                                                                                                                                                                                                                                                                                                             |

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[Setups Tab](#) : [Working with Data Streams](#) : Creating, Modifying, and Deleting a Data Stream

---

## Creating, Modifying, and Deleting a Data Stream

To create a new custom data stream, you use this section of the [Setups tab](#):

Current Data Streams

HRAM Setup: DefaultHRAMSetup

| Type        | Stream Name   | Output | Parametric Info     | Functional Info     | Sample R... | Sample Li... |
|-------------|---------------|--------|---------------------|---------------------|-------------|--------------|
| Datalog     | DatalogStream | WIN    | ALL TESTS + EXPA... | ALL TESTS + EXPA... | 1           | 0            |
| Data stream | MDSFails      | STDF   | C:\MP4\DlgSt.stp    | C:\MP4\DlgSt.stp    | 1           | 0            |
| Data stream | MDSAll        | STDF   | ALL TESTS + EXPA... | ALL TESTS + EXPA... | 1           | 0            |

New Delete

Setup Type

☒ Datalog ☐ Data Stream

☒ Window Log By Test Name :

Setups

Short Expanded Fails Only

Parametric: ☐ ☒ ☐ ☐ Customize: Setup File...

Functional: ☐ ☒ ☐ Setup File:  Browse...

Output

☐ Text File Log By Test  Browse...

☐ OverWrite File

☐ STDF File Log By Test  Browse...

☐ OverWrite File ☐ Disable Datalog Text Record

☐ Custom Filter  Browse...

Sample

Sample Rate : 1 Sample Limit : 0

☐ Enable Per-Test Sampling Select Tests ...

Display Settings... Apply Reset

## Custom Data Stream Controls

Follow these steps:

1. [Click the New button](#) under the Current Data Streams display grid. A new highlighted row appears in the grid. The Data Stream radio button is checked.
2. [Enter a name for the data stream in the Name field](#) or accept the default name.
3. [Specify a setup for the stream in the Setups frame](#), either using the controls for the built-in setups or by checking Customize for a custom setup. See [Specifying Setups](#).
4. [Select exactly one output destination for the data stream in the Output frame](#). You must check either Test File or STDF File. You can also check the Custom Filter. See [Specifying Output](#).

5. [Specify sample options in the](#) Sample frame, if desired. See [Specifying Sampling](#).
6.  [Click the](#) Apply button to save the changes to the data stream. See [Saving the Setup with Apply](#).

The new setup appears in the [Current Data Streams](#) grid.

To modify an existing custom data stream, select it in the grid. Use the controls to modify the settings. Click [Apply](#) when done.

You can use the [Reset](#) to restore the default data stream settings to the currently selected data stream. See [Resetting Data Streams](#).

To delete a custom data stream, select it in the grid and click [Delete](#). Delete is enabled only if you select a row in [Current Data Stream](#) list. If you click [Delete](#), you are prompted to confirm the permanent deletion of the stream from the list.

|                            |                                                                                                                                                                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Delete</a>     | <a href="#">Removes a data stream from the Current Data Stream list.</a>                                                                                                                                                                                                                                    |
| <a href="#">HRAM setup</a> | <a href="#">Specifies which of the created HRAM setups to apply to hardware.</a><br>DefaultHRAMsetup is included as the default HRAM setup that will be applied to the hardware. You can create more than one HRAM setup for multiple data streams.<br><a href="#">See HRAM Setup in DataCollect Setup.</a> |

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## Setups Tab : [Working with Data Streams](#) : Modifying the Built-In DatalogStream

### Modifying the Built-In DatalogStream

You can [modify the built-in](#) DatalogStream. When you click this stream in the Current Data Streams grid, the Datalog radio button under Setup Type is selected.

[The](#) DatalogStream is the only stream that can write to the OutputWin window. When the Datalog radio button is selected, the Window check-box is enabled. Check this box if you want to send datalog output to the OutputWin. If the Windows box is checked, the drop-down control next to it is enabled. This lets you select how to order the datalog output written to the window:

|                               |                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Log By Test</a>   | <a href="#">Display the collected data in order by test number.</a>                                                            |
| <a href="#">Log By Device</a> | <a href="#">Group the collected data by device number and then, for each device, display the data in order by test number.</a> |

[In addition to the](#) Window control, you can also select any other options under Output for the built-in DatalogStream. Unlike custom streams, DatalogStream can have multiple output destinations. If you select Text File or STDF File, the Log By Test/Log By Device control is enabled for that destination, as described above.

[For more information, see](#) [Specifying Output](#).

[As with custom data streams, you can also modify the](#) Setups and Sample settings for the DatalogStream. See [Specifying Setups](#) and [Specifying Sampling](#).

[You can use the](#) Reset to restore the default data stream settings to the currently selected data stream. See [Resetting Data Streams](#).

 [When you finish modifying the](#) DatalogStream settings, click Apply to save them. See [Saving the Setup with Apply](#).

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Setups Tab : [Working with Data Streams](#) : Saving the Setup with Apply

## Saving the Setup with Apply

Apply

When you have created a new data stream or modified an existing one, click the Apply button to save the changes.

If you have created or modified a data stream and then select another data stream without clicking Apply, you are asked if you want to save the modifications.

Note that the changes are not permanently saved and applied until you click the OK button. (The OK button is the equivalent of the VBT statement TheExec.Datalog.ApplySetup.) If you click Apply to save the changes to a data stream and then click Cancel to exit from the GUI, the changes to the data stream are lost. You are not asked if you want to save the changes. You must click OK to save the changes to the data stream. See [OK and Cancel](#).

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Setups Tab : Working with Data Streams : Resetting Data Streams

## Resetting Data Streams

Reset

The Reset button restores the default values to the controls for the currently selected data stream or datalog. The default values are different for the built-in DatalogStream datalog and for user-defined data streams.

For the built-in DatalogStream datalog, the following resets occur:

- The Window output option in Setup Type is checked, with the combo box set to Log By Test.
- The Expanded option is selected for both Parametric and Functional.
- Sample Rate is set to 1, the default.
- Sample Limit is set to 0, the default.
- The Enable Per-Test Sampling check box and Select Tests button are disabled.

For a user-defined data stream, the following resets occur:

- All Output selections are unchecked.
- The Name text box is set to the default data stream name, using the current system data and system time.

- The Expanded option is selected for both Parametric and Functional.
- Sample Rate is set to 1, the default.
- Sample Limit is set to 0, the default.
- The Enable Per-Test Sampling check box and Select Tests button are disabled.

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## Setups Tab : Specifying Setups

### Specifying Setups

Each data stream, including DatalogStream, has a setup that specifies what data to datalog and how to format it. Each data stream has a parametric setup and a functional setup. Use the Setups frame to specify the setup:

Setups

|             | Short                 | Expanded                         | Fails Only               |                                                                                  |
|-------------|-----------------------|----------------------------------|--------------------------|----------------------------------------------------------------------------------|
| Parametric: | <input type="radio"/> | <input checked="" type="radio"/> | <input type="checkbox"/> | <input type="checkbox"/> Customize: <input type="button" value="Setup File..."/> |
| Functional: | <input type="radio"/> | <input checked="" type="radio"/> | <input type="checkbox"/> | Setup File: <input type="text"/> <input type="button" value="Browse..."/>        |

### Setups Frame

You can use built-in setups or a customized setup.

To use built-in setups, use the Parametric and Functional controls in the Setups frame. These controls are enabled only if Customize is not checked. You can make these choices for parametric and functional tests:

|                   |                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Short or Expanded | Select either format for parametric and functional tests. See Built-In Setups (Short and Expanded).                                                  |
| Fails Only        | Specifies whether you want to display only failed tests (checked) or all tests (not checked).                                                        |
| Customize         | Specifies customized setups. The Setup File button and box are enabled. For an overview of customized setups and the setup file, see Datalog Setups. |

Each custom setup file has settings for both parametric and functional tests. When using customized setups, you can make these choices:

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browse     | To load a previously saved setup file, use this button to navigate to the file. When you save a setup file, you also save the list of setups that were then selected. When you load the file, the previously selected setups are still selected. You can use these as the currently selected setups, or you can click Setup File to bring up the Setup File dialog box to select and possibly edit the specific setups to be used. See Setup File Dialog Box. |
| Setup File | To create a new setup file, click this button to bring up the Setup File dialog box, where you can create new setups and save them to a setup file. See Setup File Dialog Box.                                                                                                                                                                                                                                                                                |

A reference to a datalog setup file that does not exist or has invalid contents returns a run-time error and stops datalog execution.

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## Setups Tab : Specifying Output

### Specifying Output

Each data stream specifies the output destination to which the datalog is to be directed.

The built-in DatalogStream can have multiple output destinations, including the OutputWin window. For more information, see [Modifying the Built-In DatalogStream](#).

A custom data stream must have exactly one output destination, either a text file or an STDF file. A custom filter can also be specified for a data stream.

The Output frame has these options:

**Output**

☒ Text File

Log By Test

☐ OverWrite File

Browse...

☒ STDF File

Log By Test

☐ OverWrite File

Browse...

☒ Custom Filter

Browse...

☐ Disable Datalog Text Record

### Output Frame

The options are as follows:

**Text File**      Write to an ASCII file. Test File has a drop-down box that lets you specify:

- **Log By Test:** Display the collected data in order by test number.
- **Log By Device:** Group the collected data by device number and then, for each device, display the data in order by test number.

Use this option to enable the field and the Browse button to specify a filename. If you do not specify a filename, a default is used.

If you check this box, the OverWrite File box is enabled. If you check this box as well, any existing file with the name you specify is overwritten.

If the same text file is specified for a datalog stream output and the summary output, an error is written the output window. The specified name is used for the summary text file, and a new name is generated for the datalog stream text file.

You cannot specify the same filename for multiple streams.

#### STDF File

Write to a binary file in STDF format.

Use this option to enable the field and the Browse button to specify a filename. If you do not specify a filename, a default is used.

For the relationship between datalog and summary STDF file output, see Output Options.

If you check this box, Disable Datalog Text Record is enabled. Check this box to exclude DTRs from the STDF file. DTRs generally contain comments to be included in the datalog printout. See [Datalog Text Record \(DTR\)](#).

If you check this box, the OverWrite File box is also enabled. If you check this box as well, any existing file with the name you specify is overwritten.

If the same STDF file is specified for a datalog stream output and the summary output, an error is written the output window. The specified name is used for the summary STDF file, and a new name is generated for the datalog stream STDF file.

You cannot specify the same filename for multiple streams.

#### Custom Filter

Write to a file in user-specified format. A .dll file is used to save the data in a customized format.

This option allows you to process test and summary data from a data stream and generate the user-customized file format or contents. Custom Filter generates the output in both text and STDF files.

For more information, see [Output Options](#).

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## Setups Tab : Specifying Sampling

### Specifying Sampling

The Sample frame contains controls to specify how the sites are to be datalogged. It applies to both datalog and data streams. The default is to datalog all sites.

The frame includes these controls:

Sample

Sample Rate : 1

Sample Limit : 2

☒ Enable Per-Test Sampling

Select Tests ...

### Sampling Frame

The options are as follows:

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sample Rate              | Specify the interval at which sites are to be sampled. Enter an integer between 1 and 1000. The default is 1, which means to datalog every site (no sampling). See Sample Rate and Sample Limit.                                                                                                                                                                                                                                                 |
| Sample Limit             | Specify the number of sites to sample. The default is 0, which means sample all sites. See Sample Rate and Sample Limit.                                                                                                                                                                                                                                                                                                                         |
| Enable Per-Test Sampling | This checkbox is enabled if Sample Rate or Sample Limit is set to a nondefault value. This feature lets you specify tests that must be datalogged for every site, even if Sample Rate specifies that only every nth site is to be datalogged. In early production or when debugging, it can be useful to collect more information about a specific parameter. This feature allows you to collect complete information for a given test or tests. |
| Select Tests             | This button is enabled if you check Enable Per-Test Sampling. Clicking the button brings up the Sampling setup dialog box, which lets you                                                                                                                                                                                                                                                                                                        |

specify which test numbers or names are to be sampled. See [Sampling Setup Dialog Box](#).

### Sample Rate and Sample Limit

This section explains how values for sample rate and sample limit are applied in a multisite program. Note that site numbering starts at 0, so a sample limit of 1 specifies site 0, while a sample limit of 2 specifies sites 0 and 1.

The sample rate value specifies the interval at which sites are to be sampled for output during each run. The default value is 1, which means sample every site during each run.

The default sample rate is 1. A sample rate of 1 means log all sites. Specifically:

- In a two-site program, results for Site0 and Site1 are logged for each run.
- In an n-site program, results for Site0, Site1, Site2, ... through to SiteN-1 are logged for each run.

A user-specified, nondefault sample rate causes only certain sites to be logged. Specifically, the first site is logged, then each nth site after is logged. For example:

- Assume a two-site program, where the sample rate is 3 and the program is looped four times. The sample rate of 3 says to log the first site (Site0), then skip the next two sites, then log the third. The result is as follows, with logged sites in bold:

– Run 1: Site0 (log), Site1 (count = 1, so skip)

– Run 2: Site0 (count = 2, so skip), Site1 (count = 3, so log)

– Run 3: Site0 (count = 1, so skip), Site1 (count = 2, so skip)

– Run 4: Site0 (count = 3, so log), Site1 (count = 1, so skip)

- Assume an eight-site program, where the sample rate is 3 and the program is looped three times. The sample rate of 3 says to log the first site (Site0), then skip the next two sites, then log the third. The result is as follows, with logged sites in bold:

- **Run 1:** Site0 (log), Site1 (count = 1, so skip), Site2 (count = 2, so skip), Site3 (count = 3, so log), Site4 (count = 1, so skip), Site5 (count = 2, so skip), Site6 (count = 3, so log), Site7 (count = 1, so skip)
- **Run 2:** Site0 (count = 2, so skip), Site1 (count = 3, so log), Site2 (count = 1, so skip), Site3 (count = 2, so skip), Site4 (count = 3, so log), Site5 (count = 1, so skip), Site6 (count = 2, so skip), Site7 (count = 3, so log)
- **Run 1:** Site0 (count = 1, so skip), Site1 (count = 2, so skip), Site2 (count = 3, so log), Site3 (count = 1, so skip), Site4 (count = 2, so skip), Site5 (count = 3, so log), Site6 (count = 1, so skip), Site7 (count = 1, so skip)

The order of logging sites applies from the first run (at the start of the test program or after an endlot) onwards. If the sample rate is changed between runs, then the order of sites will change according to run number.

The sample limit specifies the number of sites to sample. The default is 0, which means sample all sites. If the sample limit is set to 1, only the first site (Site0) is sampled. For example, assume a four-site program with a sample rate of 1 (the default - log all sites):

- If the sample limit = 0, sample all sites.
- If the sample limit = 1, sample only Site0.
- If the sample limit = 2, sample only Site0 and Site1.

If you specify a nondefault sample rate, the sample rate is applied to the sample limit value. For example, assume a four-site program:

- If the sample rate = 1 and the sample limit = 2, Site0 and Site1 are logged on the first run.
- If the sample rate = 2 and the sample limit = 2, Site0 and Site2 are logged on the first run.

In both cases, on subsequent runs or loops, there is no further datalogged output, when the sample limit is any value other than 0, the default.

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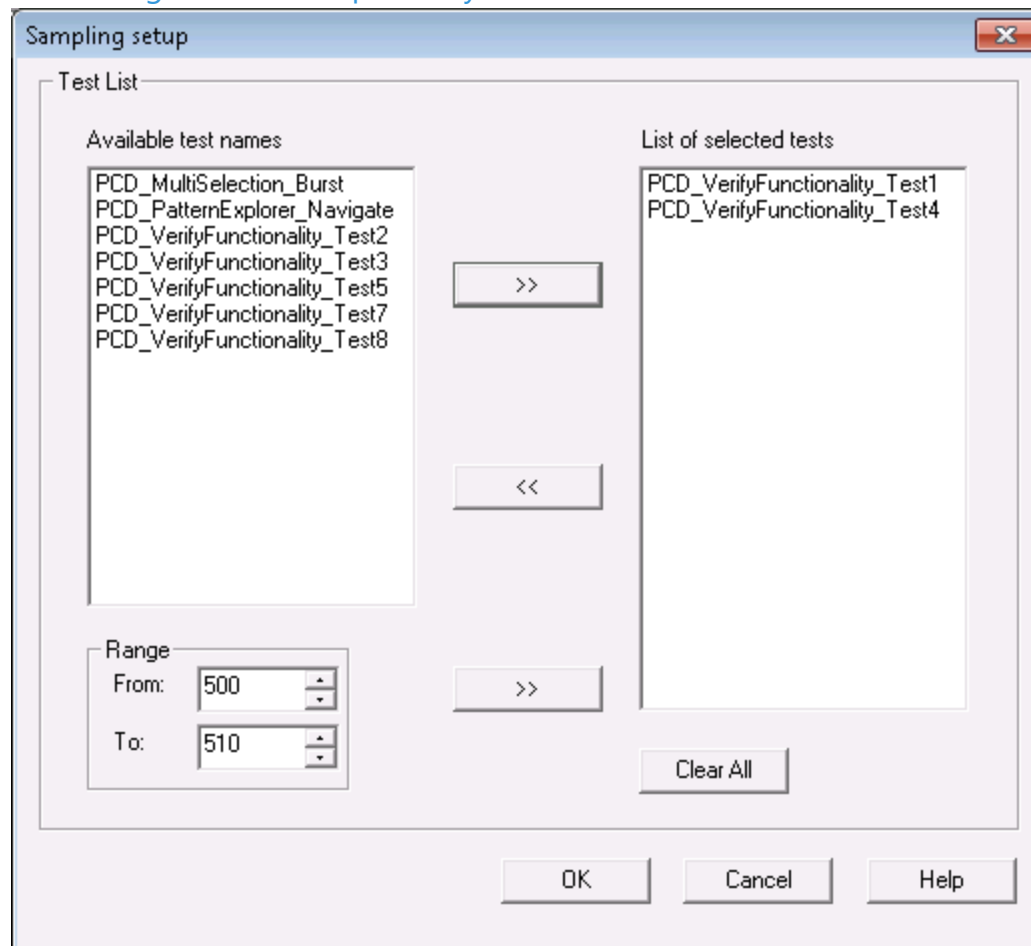
## dlsetups.2.13



Setups Tab : Specifying Sampling : Sampling Setup Dialog Box

### Sampling Setup Dialog Box

This dialog box comes up when you click Select Tests on the Lot Definition tab:



### Sampling Setup Window

The dialog box lets you create a list of tests that are always to be datalogged, regardless of the sampling rate or limit. You can specify test names and ranges of test numbers.

**Available test names** lists the test names from the Test Instances sheet. These names can be displayed only after the program workbook is validated. To select a test, highlight it and click >>.

The test name moves to the List of selected tests. To deselect a test, click <<.

[To use a test number range, use](#) From: and To:. The From number cannot be larger than the To number. Use the scrollbars or type in the number. Each number must be an integer. When the range you want is displayed, click >> next to the Range frame. The range appears in the List of selected tests. If you enter a range that overlaps an existing range, the existing range is modified to include the new range. To deselect a range, highlight it and click <<.

[To deselect the entire](#) List of selected tests, use the Clear All button. Clicking this button moves all test names back to Available test names and deletes all ranges.

[For more information, see Specifying Sampling.](#)

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## Setups Tab : General Controls

### General Controls

Most controls on the Setups tab are used to define individual data streams. Controls in this section apply to all setups:

- |   |                         |
|---|-------------------------|
| • | Enable data collection  |
| • | Following Validate/Save |
| • | Display Settings        |
| • | HRAM Setup              |
| • | OK and Cancel           |

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Setups Tab : [General Controls](#) : Enable data collection

### Enable data collection

☒ Enable data collection

[Check](#) Enable data collection to turn on datalogging. Datalogging will occur the next time you execute the test program.

[You can also use the](#) Datalog On/Off control in the Data Collect group on the IG-XL Main tab to enable or disable datalogging. Checking Enable data collection on the Setups tab immediately changes the Datalog On/Off control to On. If you change the Datalog On/Off control to On, Enable data collection is checked the next time that you open DataCollect Setup.

[You can turn off this control to stop sending the data to outputs.](#) [If you disable](#) Enable data collection, the following message is shown below the check box: All data collection outputs stopped.

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Setups Tab : [General Controls](#) : Following Validate/Save

### Following Validate/Save

Use [Following Validate/Save](#) to select the action that should be taken when you revalidate or save the program workbook during program run.

|                          |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| Following Validate/Save: | <div>Close Datalog/Discard Summary</div> <div>Close Datalog/Discard Summary</div>  |
| HRAM Setup:              | <div>Close Datalog/Write Final Summary</div> <div>Datalog/Summary Not Closed</div> |

### Following Validate/Save Control

The [VBT property](#) `TheExec.Datalog.Setup.DatalogSetup.OnValidateAction` has an enumeration that lets you select equivalent actions. See [OnValidateAction](#) in the Datalog section of the VBT Syntax Reference.

The following table lists the effects of the revalidation options.

Options for Action on Revalidation

| Option                        | Effect on Datalog, Summary, and Device Number                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GUI:                          |                                                                                                                                                                                             |
| Close Datalog/Discard Summary | <ul style="list-style-type: none"><li>Write summary to datalog output.</li></ul>                                                                                                            |
| VBT:                          |                                                                                                                                                                                             |
| tl_CloseDatalogDiscardSummary | <ul style="list-style-type: none"><li>Close datalog output files and open new ones.</li><li>Do not write summary to summary outputs.</li><li>Reset summary counters and synopsis.</li></ul> |

GUI:

Close Datalog/Write Final Summary

VB:

tl\_CloseDatalogAndSummary

- Do not reset device number to 1.
- Write summary to datalog output.
- Close datalog output files and open new ones.
- Write summary to summary outputs.
- Reset summary counters and synopsis.
- Reset device number to 1.

This option has the same effect as End Lot.

GUI:

Datalog/Summary Not Closed

VB:

tl\_DatalogAndSummaryOpen

No action. Do not:

- Write summary to datalog outputs.
- Write summary to summary outputs.
- Close datalog or summary outputs.
- Reset device number to 1.

Because revalidation can occur in different contexts, the desired effect can differ depending on the context. For example:

- You may want to revalidate in the middle of a test run. During test development, validation is needed after any change to the workbook. During production, a quick change can allow a problem program to go on running.
- You may also need to save a program workbook during a program run. The effects of saving a program are the same as revalidating.

The effects of revalidation and saving are similar to those of end lot. The potential differences are in these areas:

- Whether datalog and summary outputs are closed. end lot closes the output files and clears the filenames. For revalidation, you specify the action on the output files.
  - Whether the device number is reset. end lot resets the part number to 0. For revalidation, the action you select for the output files also determines whether the part number is reset.

The option of not closing the datalog and summary can be useful during production. For example, if a program is not yielding well, a small change could improve the yield. In this case, you would want to continue datalogging to the same files, so you would choose the “no action” option. If you make a more major change, however, you may not want this option. For example, if you change jobs and continue running, this option would cause the output files to contain data from two different jobs, with no indication that the job has changed. In this case, you may prefer to close datalog and write the final summary.

During debug, either Close Datalog/Discard Summary or Close Datalog/Write Final Summary can be useful, depending on whether you want to have summary information written to the datalog or discarded after each revalidation.

For more information, see Final Summary/End Lot and Validation.

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Setups Tab : [General Controls](#) : Display Settings

## Display Settings

Display Settings...

 The Display Settings button brings up a window for specifying what data is to be written to the datalog output and how it is to be formatted. The settings apply to all data streams. See [Display Settings](#).

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Setups Tab : [General Controls](#) : HRAM Setup

## HRAM Setup

HRAM Setup: 

DefaultHRAMSetup ▾

[The HRAM Setup button](#) lets you select a nondefault HRAM setup.

[The default setup is called](#) DefaultHRAMSetup.  
[To use a nondefault HRAM setup, you must first define it in a datalog setup file. See Datalog Setups.](#) Specifically, you create a functional setup that includes an HRAM setup. For more information, see [HRAM Setup in DataCollect Setup](#). You then assign this setup to a data stream. [The HRAM Setup drop-down list contains all setup files currently assigned to data streams. You select the setup file containing the nondefault HRAM setup that you want to use.](#)

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Setups Tab : [General Controls](#) : OK and Cancel

## OK and Cancel

OK

The OK button saves and applies all of the current settings on all tabs, not only the Setups tab. It is the GUI equivalent of the VBT method `TheExec.Datalog.ApplySetup`.  
Note that the programmatic interface through the VBT Datalog object controls many more settings than the GUI does. If you specify settings using VBT, run a job, and then bring up the [DataCollect Setup GUI](#), clicking OK may have an unexpected effect. If the programmatic settings are not supported by the GUI (that is, they do not have corresponding GUI controls), clicking OK applies only the settings that the GUI supports. This means that the programmatic settings may be lost.

Cancel

The Cancel button cancels all changes on all tabs. You are not asked if you want to save the changes.  
For more information, see [TheExec.Datalog.ApplySetup](#).

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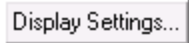
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## Setups Tab : Display Settings

---

### Display Settings

 Use this dialog box, called using the Display Settings button on the Setups tab, to specify the following output display options:

Display Options

Output Width: 1

☐ Header Every Time

☒ Test Synopsis Data

☐ Disable Channel Number in TSR Record

☐ Report By Flow Sheet Name in TSR Record

☒ Part Result

☐ X/Y coordinates

☐ Two Columns (Short Parametric)

PTR test\_txt Options

☐ Output PTR test\_txt in IMAGE format

Instance Name

☐ Disable Instance Name

☐ Test Instance Name

☒ Flow TName or Test Instance Name

☐ Disable Channel Number

☐ Disable Pin Name

☐ TSR test\_nam same as PTR test\_txt

☐ Extend MIR Job Name

☐ Full Sample STDF Part Records

☒ Append Newline after Datalog Comments in Window and Text File

OK Cancel Help

## Display Settings Window

The options are described in the following section:

- [Output Width](#)
- [Header Every Time](#)
- [Test Synopsis Data](#)

- Report By Flow Sheet Name in TSR Record

- Part Result

- Two Columns (Short Parametric)

- PTR test\_txt Options

- TSR test\_nam same as PTR test\_txt

- Extend MIR Job Name

- Full Sample STDF Part Records

- Append Newline after Datalog Comments

### Output Width

The maximum number of characters to be displayed per line. The default value is 0, which specifies no line wrapping.

### Header Every Time

Whether to repeat the report header on every test program run. If not checked, the header is printed only on the first run.

### Test Synopsis Data

Whether to include test synopsis data in the output, in addition to the datalogged data.

Test synopsis data consists of the number of failed tests and number of test executions for each site.

This box does not affect whether TSRs are written to the STDF file. TSRs are always written to the STDF file. The box affects whether the output window or text file has a per-site summary of "Failed Tests/Executed Tests".

If this box is checked, the box Disable Channel Number in TSR Record is enabled. The default format for the TSR.test\_nam record is:

test-instance-name pin-name channel-number

Checking this option causes the channel number to be removed.

### Report By Flow Sheet Name in TSR Record

Whether to use the Flow sheet name instead of the test instance name in the output to the STDF Test Synopsis Record (TSR).

The default format for the TSR.seq\_nam record is:  
test-instance-name pin-name channel-number  
Checking this option causes the test instance name to be replaced with the name of the Flow sheet in the TSR.seq\_nam record.  
For information on how IG-XL writes to the TSR, see Test Synopsis Record (TSR).

Part Result

Whether to include part result data in the output, in addition to the datalogged data.  
Part result data consists of the bin and sort numbers for each site.  
If this box is checked, the box X/Y coordinates is enabled. Checking this box causes X and Y coordinates to appear for each site after the part result in the ASCII datalog report (both window and text file). If coordinates are not set, "N/A" is displayed instead.

Two Columns (Short Parametric)

Whether to use a two-column datalog report format to condense the parametric data in the output. This option is valid when the parametric report uses the Short or Short+PinName format. (The Short format is an option for standard parametric setups. Short+PinName is an option for customized parametric setups.)

PTR test\_txt Options

Several checkboxes control the contents of the test\_txt field of the STDF Parametric Test Record (PTR).  
Output PTR test\_txt in IMAGE format specifies whether to put the items in the test\_txt field in the order used by Teradyne's IMAGE tester executive. The default format for the PTR.test\_txt record is:  
test-instance-name pin-name channel-number <> comment  
where comment is the value of the pComment parameter to the VBT function WriteParametricTestResult.  
With this option checked, the IMAGE-style format is:  
comment <> test-instance-name pin-name channel-number  
If the parametric setup specifies the Short format, the option of using IMAGE format is also available. In short format, this field always includes the test instance name and never includes the channel number or pin name. For more information, see Built-In Setups (Short and Expanded).  
The Instance Name section controls whether and how to include the instance name in the test\_txt field. If Disable Instance Name is checked, the instance name is not included. If it is not checked, the radio buttons beneath it are enabled. Use these buttons to specify which instance name to use:

|                                  |                                                                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Instance Name               | Use the name of the test instance.                                                                                                                |
| Flow TName or Test Instance Name | Use the test name from the Flow, if one is specified in the TName column of the Flow Table. If no TName is specified, use the test instance name. |

Two other checkboxes let you individually disable items that are part of both the default format and the IMAGE format:

- [Disable Channel Number](#)

- [Disable Pin Name](#)

For information on how IG-XL writes to the PTR, see [Parametric Test Record \(PTR\)](#).

TSR test\_nam same as PTR test\_txt

Whether to use the STDF Parametric Test Record (PTR) test\_txt record for the Test Synopsis Record (TSR) test\_nam record.

The default format for the TSR.test\_nam record is:

test-instance-name pin-name channel-number

Checking this option causes the TSR.test\_name record to look like the PTR.test\_txt record, which has this default:

test-instance-name pin-name channel-number <> comment

Note that this option interacts with the [Output PTR test\\_txt in IMAGE format, Report By Flow Sheet Name in TSR Record](#), and [Disable Channel Number in TSR Record options](#).

If the [parametric setup specifies the Short format](#), the option of using the nondefault format is also available. In short format, this field always includes the test instance name and never includes the channel number or pin name. See [Built-In Setups \(Short and Expanded\)](#).

For information on how IG-XL writes to the TSR and PTR, see [Test Synopsis Record \(TSR\)](#) and [Parametric Test Record \(PTR\)](#).

[Extend MIR Job Name](#)

Whether to append the IG-XL job name to the Master Information Record (MIR) job\_nam field in the STDF output.

By default, the MIR.job\_nam field uses the IG-XL program name:

ProgramName.xls

Checking this option appends the name of the current Job sheet, if there is one:

ProgramName.xls@jobname

If there is no Job sheet, the default MIR.job\_nam is used.

For information on how IG-XL writes to the MIR, see [Master Information Record \(MIR\)](#).

[Full Sample STDF Part Records](#)

When sample datalogging is being used, for a sample rate of n, every nth part is datalogged. By default, a PIR/PRR pair for a part is written only if the part is datalogged. In addition, a BPS/EPS pair for a test run is written only if parts are datalogged for that run.

This box, if checked, overrides the default behavior:

- [A PIR/PRR pair is logged for every part, whether or not the part is datalogged.](#)
- 
- [A BPS/EPS pair is logged for each test run, even if there are no parts datalogged in the test run.](#)

For more information on STDF records, see:

- [STDF Record Types](#)
- 
- [IG-XL Support for STDF Record Types](#)

[Append Newline after Datalog Comments](#)

[Whether to append a newline after datalog comments in the window and text file. By default, this box is checked, and a newline character is appended to the comment written by TheExec.Datalog.WriteComment. Unchecking this box means that a newline will not be appended. The full name of this option is Append Newline after Datalog Comments in Window and Text File.](#)

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## Datalog Setups

### Datalog Setups

A datalog setup specifies what information should be datalogged and how datalogged information should be displayed for a set of parametric or functional tests. Parametric and functional tests have different setup parameters.

The Setups tab gives you these options:

- You can choose a built-in simple setup: short or expanded format, for all tests or fails only. You can choose one setup for parametric tests and one for functional tests. See [Built-In Setups \(Short and Expanded\)](#).
- You can create customized setups and save them to a setup file. A setup file can contain any number of individual named setups for parametric and functional tests. You can load this file and select the specific setups you want to apply to this test run.

Controls for choosing standard setups are described with the [Setups tab](#). See [Specifying Setups](#). This section focuses on customized setups and the setup file:

- [Overview of Setups and the Setup File](#)
- [Built-In Setups \(Short and Expanded\)](#)
- [Setup File Dialog Box](#)
- [Conflicting or Overlapping Setups](#)

- [Parametric Setup Dialog Box](#)
- [Functional Setup Dialog Box](#)
- [HRRM Setup in DataCollect Setup](#)

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[Datalog Setups](#) : Overview of Setups and the Setup File

Overview of Setups and the Setup File

This section provides an overview of datalog setups. It includes the following topics:

- |   |                                           |
|---|-------------------------------------------|
| • | <a href="#">Datalog Information</a>       |
| • | <a href="#">Customized Datalog Setups</a> |
| • | <a href="#">Datalog Setup File</a>        |
| • | <a href="#">Using Datalog Setups</a>      |

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Datalog Setups : Overview of Setups and the Setup File : Datalog Information

Datalog Information

Datalog data gives the result of each test execution, including:

- |  |                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>Header information, including lot ID, operator, and test time</li></ul>                                                                                                                               |
|  | <ul style="list-style-type: none"><li>For parametric tests, the test name and result, with optional inclusion of test limits (see Parametric Setup Dialog Box)</li></ul>                                                                    |
|  | <ul style="list-style-type: none"><li>For functional tests, the test name and result, with optional specification of which pins to show, optional detail such as timing set, flags, and opcodes (see Functional Setup Dialog Box)</li></ul> |
|  | <ul style="list-style-type: none"><li>Test synopsis data (optional): for each site, number of failed tests and number of test executions (see Display Settings)</li></ul>                                                                   |
|  | <ul style="list-style-type: none"><li>Part result data (optional): for each site, the sort number and bin number (see Display Settings)</li></ul>                                                                                           |

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## Datalog Setups : Overview of Setups and the Setup File : Customized Datalog Setups

### Customized Datalog Setups

A datalog setup is a named set of datalogging specifications that applies to a set of tests. A setup provides details about what data to display for each test and which tests this setup should apply to. Each setup contains the test criteria defining the set of tests this setup should apply to: for example, all tests, or failed tests, or a list of specific tests.

Parametric and functional tests have separate display parameters, containing different information about how to display the data.

When you create a setup, you assign a name to it, which you use when you select setups to be used for a specific testing session.

You can select more than one setup to be used in a testing session. This means that different kinds of tests can be displayed with different information. For example, a setup for failed tests can display full detail, and another setup for passed tests can display less detail.

When you select setups to use in a testing session, two or more setups may overlap. For example, you may have one parametric setup for failed tests and another for a list of tests that includes test 100. If test 100 fails, you must be able to determine which setup applies to test 100. See [Conflicting or Overlapping Setups](#).

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## Datalog Setups : Overview of Setups and the Setup File : Datalog Setup File

### Datalog Setup File

When you create datalog setups, you can save them to a setup file, which has a .stp extension. A setup file can contain any number of named parametric and functional datalog setups. In a later session, you can load the setup file and select the names of the setups that you want to use for this test run. You can also edit setups and save them again.

For example, a setup file can contain setups that you ordinarily use for testing, plus more rarely used setups that are for debugging. A file can contain contradictory setups. For example, you can have two parametric setups for failed tests. You choose one of them (at most) only when you actually select the setups to be used in a testing session.

Although a setup file is an ASCII file, do not directly edit it. Use DataCollect Setup to create and edit setups.

If you save a setup file, you also save the list of setups that are currently selected. When you later load the file, the previously selected setups are selected. If you make changes to the selections, these changes are recorded in the setup file.

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## Datalog Setups : Overview of Setups and the Setup File : Using Datalog Setups

### Using Datalog Setups

To use datalog setups, follow these steps:

- [Validate the test program before you create or edit datalog setups. After validation, test and pin names become available in the setup dialog boxes.](#)
- [For the built-in Short and Expanded setups, use the controls on the Setups tab \(see \[Specifying Setups\]\(#\)\) to select the built-in setup for parametric and functional tests.](#)
- [For customized setups, click](#) Customize on the Datalog Setup tab.
- [To create a new setup, click](#) New. A new setup is created with a default name.
- [To use an existing setup file:](#)
  - [Use the](#) Browse button to enter its name in Setup File;; otherwise, you will create a new setup file.
  - [Then click](#) Setup File. This brings up the Setup File dialog box (see [Setup File Dialog Box](#)), which lists available and selected setups for both parametric and functional tests.
  - [Use the arrows to move existing setups into or out of the](#) Selected Setups lists.

- [To edit a setup:](#)
  - [Select it by clicking it \(in either the Available or the Selected window\).](#)
  - [Select a format by selecting one of the Format options.](#)
  - [You can also select the criteria for the tests to which this setup should apply by selecting from the drop-down list.](#)
  - [For more control over the format or criteria in the setup, click Full Edit or double-click the setup name. This brings up the Parametric Setup dialog box \(see \[Parametric Setup Dialog Box\]\(#\)\) or the Functional Setup dialog box \(see \[Functional Setup Dialog Box\]\(#\)\) which permit full editing of setups.](#)
- [Click Setup File to set up HRAM collection for functional tests. The Setup File dialog box lets you specify key HRAM parameters. To edit other HRAM parameters, click Full Edit, which brings up the HRAM Setup dialog box \(see \[HRAM Setup in DataCollect Setup\]\(#\)\).](#)
- [After creating, editing, and selecting setups, the Setup File dialog box lets you save the current settings to a setup file. If the currently selected setups violate the uniqueness requirements for the test criteria \(see \[Conflicting or Overlapping Setups\]\(#\)\), you cannot save the file until you meet the requirements.](#)
- [In a later session, when you want to use the setups, you can load the saved file:](#)
  - [On the Setups tab, select Customize and click Browse to select.](#)
  - [Under Setups, check Setup File, and then use the Browser to specify the file.](#)
  - [To select from the datalog setups, or to edit existing setups or create new ones, click Edit. Datalog Setup File appears.](#)

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## Datalog Setups : Built-In Setups (Short and Expanded)

### Built-In Setups (Short and Expanded)

The Setups tab lets you select standard setups for parametric and functional datalogging. You can also create a customized setup. This topic describes standard setups. It includes the following information:

- Short and Expanded Formats
- Test Instance Name
- Parametric Formats
- Functional Formats

To create a customized setup, see the [Setup File Dialog Box](#).  
For more information, see [Overview of Using the Setups Tab](#).

### Short and Expanded Formats

Standard datalog setups let you select between two datalog report formats: Short and Expanded. Select one format for parametric tests and one for functional tests. The meanings of the formats differ for parametric and functional tests.

For parametric tests, Short format includes: Number, Site, Test Name, and Measured. Expanded format adds: Pin, Channel, Low, High, Force, and Loc.

For functional tests, Short format includes: Number, Site, Test Name, Pattern, 1stFailVec, and TotalFailVec. Expanded format adds: pin name, the cycle number, the relative cycle number, the repeat ("Rep"), and the value of each pin for each failed cycle.

If you select a standard format, you can use [Fails Only](#) to specify the tests to datalog:

- All tests (not checked)
- Only fails (checked)

Note that these controls are disabled if Customize is checked. You can use either the standard setups or customized setups, but not both.

Test Instance Name

For all formats, the test instance name is displayed in angle brackets (for example, "<Continuity>") on the line preceding the instance's datalogged results.

Parametric Formats

The Short format for parametric tests displays four data fields for each test, with these column headings:

|           |                                                                          |
|-----------|--------------------------------------------------------------------------|
| Number    | Test number                                                              |
| Site      | Site number                                                              |
| Test Name |                                                                          |
| Measured  | Measured value. If a test fails, a red "(F)" follows the measured value. |

Note that the short parametric format permits a two-column display format. If more than one device is tested, the results for a single device appear in one column, and for the next device in the other column. To request this option, click Display Settings on the Setups tab. A control on the Display Options dialog box enables and disables the two-column format.

The Expanded format for parametric tests displays 10 data fields for each test. In addition to the four shown in the short format, the following are also shown, with these column headings:

|         |                |
|---------|----------------|
| Pin     | Pin name       |
| Channel | Channel number |
| Low     | Low limit      |
| High    | High limit     |
| Force   | Force value    |

Loc IG-XL sets this column to 0. The methods `TheExec.Datalog.WriteParametricResult` and `WriteParametricResultOptLoHi` have a parameter that lets you write a Long value to this column.

Functional Formats

The Short format for functional tests displays data fields for each test, with these column headings:

|              |                                                             |
|--------------|-------------------------------------------------------------|
| Number       | Test number                                                 |
| Site         | Site number                                                 |
| Test Name    |                                                             |
| Pattern      | If the pattern fails, a red "(F)" follows the pattern name. |
| 1stFailVec   | First failed vector                                         |
| TotalFailVec | Total number of failed vectors                              |

The Expanded format for functional tests displays more information for failed tests. For passing tests, it uses the short format. For failed tests, it displays the six fields of the short format, plus the pin name, the cycle number, the relative cycle number, the repeat ("Rep"), and the value of each pin for each failed cycle. (Failing cycles have "->FAIL" at the beginning of the line.)

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## Datalog Setups : Setup File Dialog Box

### Setup File Dialog Box

The Setup File dialog box lets you select or edit existing datalog setups or create new ones. You can make simple edits to the setups or go to another dialog box to make a full edit. You can also create an HRAM setup. You can then save the setups to a setup file.

To invoke this dialog box:

1. [Select](#) Customize on the Setups tab.

2. [Click](#) Setup File.

If a .stp file appears in the Setup File: box, that file is opened for editing from this Setup File dialog box. If no filename appears in the box, a new setup file (Untitled.stp) is created for editing from this dialog box.

For more information, see [Overview of Using the Setups Tab](#).

### Available Setups and Selected Setups Windows

Select setups separately for parametric and functional tests. Only setups in the Selected Setups window are used for current datalogging. Setups in the Available Setups window are stored in the setup file but are not used for datalogging until they are moved to the Selected Setups window.

If you load an existing setup file on the Setups tab, its setups appear in these windows. The setups that were available when the file was saved appear in the Selected Setups windows. If you did not load an existing setup file, all windows are empty.

You have these options for working with datalog setups:

|                           |                                                                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To select a setup         | Highlight the setup in the Available Setups window and click the > > button. This moves the setup to the Selected Setups window.                                                                  |
| To de-select a setup      | Highlight the setup in the Selected Setup window and click the < < button. This moves the setup to the Available Setups window.                                                                   |
| To delete a setup         | Highlight the setup in the Available Setups window and click the Delete button. You cannot delete a setup from the Selected Setups window. You must first move it to the Available Setups window. |
| To create a new setup     | Click New. A new setup is added to the Selected Setups window, with the name ParaSetup<n> or FuncSetup<n>. You can now edit the setup (including the name).                                       |
| To edit an existing setup | Select it. You can edit a setup in either the Available Setups or the Selected Setups window.                                                                                                     |

For simple edits, use the controls to the right of the windows to change the format and criteria for the setup. For the meaning of the options, see [Functional Setup Dialog Box](#) and [Parametric Setup Dialog Box](#).

For more complex edits, click Full Edit. This brings up the Functional Test dialog box (see [Functional Setup Dialog Box](#)) or the Parametric Test dialog box (see [Parametric Setup Dialog Box](#)) for editing the currently selected setup. These dialog boxes contain all controls for editing setups, including the simple controls shown on the Setup File dialog box.

Note that some simple editing choices may require use of the full edit dialog box. For example, if you select a criterion that uses a test list, you must use the full edit dialog box to specify the list.

#### Conflicting or Overlapping Setups

When you are selecting setups, two selected setups can apply to the same test. Some conflicts prevent you from exiting from this dialog box or saving the setups to a file. See [Conflicting or Overlapping Setups](#) on resolving such conflicts.

#### HRAM

The lower section of the dialog box sets up HRAM (History RAM) for functional tests. Use the controls in this section to set up the most commonly modified HRAM items. You can also click Full Edit to edit all HRAM items.

The HRAM Setup dialog box lets you either apply the HRAM setup to all time domains or create a separate setup for each time domain. The options available here on the Setup File dialog box apply to all time domains or to a single time domain, if only one is used. If you use the HRAM Setup dialog box to create different HRAM setups for different time domains, the HRAM controls here on the Setup File dialog box are disabled.

For more information, see [HRAM Setup in DataCollect Setup](#).

#### Save and Save As

Click the Save button to save any changes to the setup file you have loaded.

Click the Save As button to save the current setups to a new setup file. You can browse to the new filename.

If you did not load a file, you cannot save to the default filename "Untitled.stp." In this case, Save functions as Save As, and you must specify a new filename.

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## Datalog Setups : Conflicting or Overlapping Setups

### Conflicting or Overlapping Setups

Each datalog setup specifies the criteria for the tests to which it applies. For example, a parametric test setup can specify all tests or a list of tests.

When you select setups in the Setup File dialog box, a given test can appear in the specified criteria of more than one setup. For example:

- If one setup has All tests and another has Failed tests, a test that fails is also included in All tests.
- Two setups can both specify All tests.

Note that parametric setups and functional setups are considered separately. For example, All tests in parametric setups does not conflict with All tests in functional setups.

If you select setups that have overlapping criteria, IG-XL resolves conflicts using the rules described here.

For more information, see Setup File Dialog Box.

### Uniqueness Requirement

When you select either parametric or functional setups, each of these criteria can appear in only one setup:

- Failed tests
- Passed tests

- All tests

That is, only one of the selected parametric setups can have Failed tests as its criteria. You can, however, have one parametric setup with Failed tests and one with All tests. Any overlap is resolved by the next rule.

If two or more setups in the Selected Setups window of the Setup File tab have the same criteria, you cannot click OK to exit from the tab. Clicking OK displays an error message until you deselect setups to make the criteria unique.

#### Resolving Overlapping Criteria

Apart from the uniqueness requirement, any combination of criteria is possible, and criteria from the selected setups can overlap. To resolve overlapping criteria, IG-XL uses this list to determine priority:

- Failed in List

- Passed in List

- Failed + List

- Passed + List

- List

- Failed tests

- Passed tests

- All tests

If a test appears in the criteria of two different setups, the setup that appears higher in the list is used for the test. For example, if you select a setup with Failed tests and one with All tests, a failed test will use the setup with the Failed tests criteria, because it is higher in the list than All tests.

If two setups have the same criteria (which is permitted, except for the three criteria listed above under the uniqueness requirement) - for example, Failed in List - and if a test appears in both criteria, the criteria have the same priority. In this case, the setup that appears earlier in the list of selected setups is applied.

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## Datalog Setups : Parametric Setup Dialog Box

### Parametric Setup Dialog Box

The Parametric Setup dialog box lets you edit a datalog setup for a set of parametric tests. Invoke this dialog box from the Setup File dialog box by highlighting a setup and clicking Full Edit. The simple editing controls that appear on the Setup File dialog box for specifying the Format and Criteria for the setup also appear on this Parametric Setup dialog box.

The following topics are included:

- |   |           |
|---|-----------|
| • | Name      |
| • | Format    |
| • | Criteria  |
| • | Test List |

For more information, see:

- |   |                       |
|---|-----------------------|
| • | Datalog Setups        |
| • | Setup File Dialog Box |

### Name

The name for this setup. Enter a string, which can include blanks. Use this box to change the default name given to a setup.

Each parametric test setup must have a unique name, although a parametric test setup can have the same name as a functional test setup.

Format

What data to display for each datalogged parametric test. The options are:

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Short (test name and result)                     | The standard short format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Short2 (test name and result (1 pass line/site)) | <p>The standard short format, but displaying only one pass result for each site. This format reduces the number of lines output by displaying only the first passing pin (with the pin number) and omitting all the other passing pins. A separate line is still printed for each failing pin.</p> <p><b>Note that</b> TheHdw.PPMU.Pins.Test logs exactly one result per site. This is the result of all pins per site. If all pins pass, the result is pass; if one or more pins fail, the result is fail. The PPMU test from the VBT test library uses this method and therefore produces the same output.</p> |
| Short+PinName (test name, result and pin name)   | <p>The standard short format, plus an additional column Pin Name.</p> <p>The pin name is written only to the text and window outputs. This short format does not write the pin name to the STDF PTR. In Expanded format, you can have the pin name written to the TEST_TXT field of the PTR: see PTR test_txt Options.</p>                                                                                                                                                                                                                                                                                       |
| Expanded (test name, result and limits)          | The standard expanded format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

All options except Short2 display one line of data for each pin tested at a site. On short and expanded formats, see Built-In Setups (Short and Expanded). Note that you can apply the two-column parametric format to any of the short formats listed here. Specify the two-column format in the Display Options dialog box.

Criteria

The parametric tests to which this setup should apply. The options are:

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| All tests     | All tests.                                                                        |
| Passed tests  | All passed tests.                                                                 |
| Failed tests  | All failed tests.                                                                 |
| List of tests | A list of test numbers or names created in the Test List frame of the dialog box. |

|                |                                                               |
|----------------|---------------------------------------------------------------|
| Passed + List  | All passed tests plus all tests in the list you create.       |
| Failed + List  | All failed tests plus all tests in the list you create.       |
| Passed in List | Only the passed tests among the tests in the list you create. |
| Failed in List | Only the failed tests among the tests in the list you create. |

Test List

This frame allows you to create a list of tests to be used for the List criteria. It is enabled only if the selected Criteria includes List. You can specify test names and ranges of test numbers.

Available Tests lists the parametric test names from the Test Instances sheet. These names can be displayed only after the program workbook is validated. To select a test, highlight it and click > >. The test name moves to the List of Selected Tests. To deselect a test, click < <.

To use a test number range, use From: and To:. The From number cannot be larger than the To number. Use the scrollbars or type in the number. Each number must be an integer. When the range you want is displayed, click > > next to the Range frame. The range appears in the List of Selected Tests. If you enter a range that overlaps an existing range, the existing range is modified to include the new range. To deselect a range, highlight it and click < <.

To deselect the entire List of Selected Tests, use the Clear button. Clear moves all test names back to Available Tests and deletes all ranges.

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## Datalog Setups : Functional Setup Dialog Box

### Functional Setup Dialog Box

The Functional Setup dialog box lets you edit a datalog setup for a set of functional tests. Invoke the dialog box from the Setup File dialog box by highlighting a setup and clicking Full Edit. The controls that appear on the Setup File dialog box for specifying the Format and Criteria for the setup also appear on this Functional Setup dialog box.

The following topics are included:

- |   |           |
|---|-----------|
| • | Name      |
| • | Format    |
| • | Criteria  |
| • | Test List |
| • | Pins      |
| • | Display   |

For more information, see:

- |   |                       |
|---|-----------------------|
| • | Datalog Setups        |
| • | Setup File Dialog Box |

Name

The name for this setup. Enter a string which can include blanks. Use this box to change the default name given to the setup.

Each functional test setup must have a unique name, although a functional test setup can have the same name as a parametric test setup.

Format

What data to display for each datalogged functional test. The options are:

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Short (test name and result)            | The standard short format.                                                                                          |
| Expanded (test name, result and limits) | The standard expanded format. Enables the Display and Pins frames so you can specify further details of the format. |

For more information, see Built-In Setups (Short and Expanded).

Criteria

The functional tests to which this setup should apply. The options are:

|                |                                                                                   |
|----------------|-----------------------------------------------------------------------------------|
| All tests      | All tests.                                                                        |
| Passed tests   | All passed tests.                                                                 |
| Failed tests   | All failed tests.                                                                 |
| List of tests  | A list of test numbers or names created in the Test List frame of the dialog box. |
| Passed + List  | All passed tests plus all tests in the list you create.                           |
| Failed + List  | All failed tests plus all tests in the list you create.                           |
| Passed in List | Only the passed tests among the tests in the list you create.                     |
| Failed in List | Only the failed tests among the tests in the list you create.                     |

Test List

This frame allows you to create a list of tests to be used for the List criteria. It is enabled only if the selected Criteria includes List. You can specify test names and ranges of test numbers.

Available Tests lists the parametric test names from the Test Instances sheet. These names can be displayed only after the program workbook is validated. To select a test, highlight it and click > >. The test name moves to the List of Selected Tests. To deselect a test, click < <.

To use a test number range, use From: and To:. The From number cannot be larger than the To number. Use the scrollbars or type in the number. Each number must be an integer. When the range you want is displayed, click >> next to the Range frame. The range appears in the List of Selected Tests. If you enter a range that overlaps an existing range, the existing range is modified to include the new range. To deselect a range, highlight it and click <<.

To deselect the entire List of Selected Tests, use the Clear button. Clear moves all test names back to Available Tests and deletes all ranges.

Pins

This frame allows you to select which pins should be displayed. This frame is enabled only if Format is Expanded.

The options are:

- Failing pins
- All pins
- List of pins
- Failing pins in list

The last two options use a list of pins that you create in Available Pins and List of Selected Pins. Available Pins lists the pins from the Pin Map sheet. These names can be displayed only after the program workbook is validated. To select a pin, highlight it and click >>. To deselect a pin, click <<. Use the Clear button to deselect all pins.

Display

This frame includes other display options for the datalog. It is enabled only if Format is Expanded. The options are:

|                      |                                                                                                                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pattern column order | Display individual pins in the order in which they appear in the pattern without displaying groups.                                                                                                                                                                                                   |
| Pattern column group | Display pin groups in the order in which they appear in the pattern. This item is enabled only if you chose All pins in the Pins frame. If you select Pattern column group, Channel Numbers and Physical Names are disabled (these cannot be used with pin groups) and Space Between Pins is enabled. |
| Pin Names            | Display pin names for each test execution.                                                                                                                                                                                                                                                            |

|                    |                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Channel Numbers    | Display channel numbers for each test execution. This item is enabled only if Pattern column order is selected.                                                                               |
| Physical Names     | Display physical names for each test execution. Physical names are taken from the Package Pin column of the Channel Map sheet. This item is enabled only if Pattern column order is selected. |
| Timing Sets        | Display timing sets for each test execution.                                                                                                                                                  |
| Flags              | Display flags for each test execution.                                                                                                                                                        |
| Scan Vectors       | Display scan vectors for each test execution.                                                                                                                                                 |
| Opcodes            | Display opcodes for each test execution.                                                                                                                                                      |
| Labels             | Display labels for each test execution.                                                                                                                                                       |
| Space Between Pins | Put space between the pin groups for the pin data. This item is enabled only if Pattern column group is selected.                                                                             |

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## Datalog Setups : HRAM Setup in DataCollect Setup

### HRAM Setup in DataCollect Setup

History RAM (HRAM) is a pattern memory that captures data about pattern cycles while the pattern is executing in a functional test. After execution, the data about each captured cycle is available for viewing.

An HRAM setup includes the following kinds of information:

|              |                                                                                 |
|--------------|---------------------------------------------------------------------------------|
| Time domains | Whether the HRAM setup applies to all time domains or to a specific time domain |
| Trigger      | The point in the pattern where capturing is to begin.                           |
| Capture      | What cycles are to be captured after the trigger is encountered.                |

The HRAM Setup dialog box lets you specify how HRAM should be triggered and captured during the current testing session. Invoke the dialog box from the Setup File dialog box (see [Setup File Dialog Box](#)) by clicking Full Edit in the HRAM frame. The simple HRAM setup controls that appear on the Setup File dialog box also appear on this HRAM Setup dialog box.

The following topics are included:

- When HRAM Setup Is Applied
- Time Domains
- Trigger Frame

- [Capture Frame](#)
- 
- [HRRM Reburst](#)
- 
- [Defaults for HRRM](#)

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## Datalog Setups : HRAM Setup in DataCollect Setup : When HRAM Setup Is Applied

### When HRAM Setup Is Applied

The HRAM setup you specify is applied to the hardware the next time a functional test runs a pattern. Specifically, the HRAM setup is applied to the hardware when the functional test calls the [VBT Pattern object's method .Test method](#). The method applies the HRAM setup before it starts the test pattern.

Note that the HRAM Setup dialog box does not display the current hardware HRAM setup. Note also that the setup is not applied to the hardware when you click OK.

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## Datalog Setups : HRAM Setup in DataCollect Setup : Time Domains

### Time Domains

The HRAM Setup dialog box lets you either apply the HRAM setup to all time domains or create a separate setup for each time domain. If you use the HRAM Setup dialog box to create an HRAM setup for all time domains, then the HRAM controls are enabled. If you use the HRAM Setup dialog box to create different HRAM setups for different time domains, then the HRAM controls on the Setup File dialog box are disabled.

At the top of the HRAM Setup dialog box is a checkbox labelled Use All Domains Setup. If you check this box, the HRAM setup applies to all time domains. If you uncheck the box, the HRAM setup applies only to the time domain selected in the Time Domain field, and the HRAM controls on the Setup File dialog box are disabled.

If this setup applies to only one time domain, use the Time Domain pull-down menu to specify the time domain to which it applies. The pull-down list contains the name of all time domains defined on the Pin Map sheet. The list also contains an entry for All Domains. This option lets you define a setup that applies to all domains and then define separate setups for individual time domains. The All Domains setup acts as a default that is overridden by a setup defined for an individual time domain.

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## Datalog Setups : HRAM Setup in DataCollect Setup : Trigger Frame

### Trigger Frame

The Trigger frame specifies the point in the pattern where the trigger is to occur. Trigger controls are as follows:

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| On First:        | <p>Specifies the cycle on which HRAM capture trigger is to occur:</p> <ul style="list-style-type: none"><li>• <b>Cycle</b>: Trigger on the very first cycle in the pattern.</li><li>• <b>Fail</b>: Trigger on the first cycle that fails.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| After cycle with | <p>Specifies the conditions that must be satisfied before triggering occurs. Use this item to delay the trigger specified as On First: until after this point:</p> <ul style="list-style-type: none"><li>• <b>Cycle Number</b>: Start after the executed cycle count reaches the number you specify.</li><li>• <b>Vector</b>: Start after the vector with the label you specify, in the pattern you specify. (In Pattern is enabled only if you check Vector.)</li><li>• <b>Loop/Repeat</b>: Start after the loop counter reaches the number you specify. The counter to be monitored is the one at the top of the stack for the given cycle: for example, if a given cycle corresponds</li></ul> |

to a repeat inside a loop, the repeat counter is at the top of the stack and is therefore the one to be monitored.

Once these conditions are satisfied, capturing is triggered on the first cycle that meets the condition specified by On First. If you enable one or more of these items, all of the enabled conditions must be satisfied on the same cycle to enable triggering. This permits complex triggering, such as: "Trigger on the first failure after the pattern reaches vector label 'loop1' and the loop counter equals 56."

Stop On Full

This option is always checked and disabled. HRAM capturing always stops after it reaches its size limit and is about to overwrite data that has already been captured.

Pretrigger Cycles

Specifies the number of captured cycles to keep in HRAM just before the trigger cycle. Pretrigger is useful for capturing enough cycles to see the context of a failure.

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## Datalog Setups : HRAM Setup in DataCollect Setup : Capture Frame

### Capture Frame

The Capture frame specifies what cycles are to be captured. The controls are as follows.

|                   |                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capture           | <p>Selects which vectors are to be captured once capturing is triggered:</p> <ul style="list-style-type: none"><li>• All: Capture all vectors.</li><li>• Fail: Capture failing vectors.</li><li>• STV: Capture vectors containing the control bit stv (Store This Vector).</li></ul>                                                                                                  |
| Memory Size       | <p>Sets the memory size, in number of cycles. This item makes the size of HRAM memory size programmable in software.</p> <p>You can set a size smaller than the hardware size for efficiency. You can try to reduce the size to the minimum necessary to reduce readback time.</p> <p>A size of zero (0) has a special meaning: Use the maximum number that will fit in hardware.</p> |
| Compress repeats  | <p>This option is always unchecked and disabled. The HRAM hardware does not support this option.</p>                                                                                                                                                                                                                                                                                  |
| Capture keepalive | <p>This option is always checked and disabled. HRAM always captures keepalive.</p>                                                                                                                                                                                                                                                                                                    |

|                   |                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------|
| Loop select       | This option is always disabled. See the description of After cycle with Loop/Repeat under the Trigger frame. |
| Reburst           | Enables the HRAM reburst capability. See HRAM Reburst.                                                       |
| Refresh HRAM data | Refreshes HRAM data after each capture.                                                                      |

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Datalog Setups : [HRAM Setup in DataCollect Setup](#) : HRAM Reburst

## HRAM Reburst

[Checking the](#) Reburst box in the Capture frame enables the HRAM reburst capability, which lets you capture more cycles than fit in the hardware HRAM. This feature causes IG-XL to reburst the PatGen, using a sliding window of captured cycles for each reburst, as many times as is needed to capture the number of cycles requested. The feature lets you datalog all cycles executed in a pattern burst without being limited to the size of the hardware HRAM.

[HRAM reburst is most typically used to capture only failing cycles, but it can be used with any HRAM capture mode.](#)

[After reburst, you can use the VBT language object TheHdw.Digital.HRAM to examine the contents of HRAM and to determine such information as the number of captured cycles. Note that the HRAM VBT statements return only data that is currently in HRAM, that is, from the last PatGen burst that occurred. For HRAM reburst, the HRAM contains only the last window of data. Earlier reburst data is not available from VBT. For more information, see TheHdw.Digital.HRAM.](#)

[Note these restrictions and cautions on using HRAM reburst:](#)

- [Any programmed PatGen or trigger events are valid only for the first PatGen burst. Subsequent bursts reset these events to capture the requested amount of data. The original HRAM setup is not restored after the data is captured.](#)
- [For HRAM reburst to work, only one site can be active. If more than one site is active, an error is returned.](#)
- [If you have a prepattern or postpattern interpose function, it will be called for each PatGen reburst that is required to collect all of the data. If the interpose function modifies the HRAM setup \(for example, by changing the patgen event\), the reburst may not function correctly.](#)

- The reburst feature requires the PatGen burst to be repeatable. If a different number of PatGen cycles is executed for each burst, the algorithm to collect the data will not work. Therefore, if portions of a pattern will execute an indeterminate number of cycles (for example, a match loop or waiting for a PatGen flag), the cycle counter must be disabled so that the algorithm will work. Use the `disable_sync_ctr` opcode to disable counting cycles during these portions of the pattern. This may result in the indeterminate cycles not being datalogged.

Another problematic case is when you are capturing only fails and a different number of fails occurs on each PatGen burst. For example, a fail may occur the first time the PatGen is run, but not on the next. Therefore, some fails may not be datalogged. This could cause the fail count displayed at the top of the datalog not to agree with the detailed information printed below: the fail count could say that five fails occurred, but only four fails are displayed.

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Datalog Setups : HRAM Setup in DataCollect Setup : Defaults for HRAM

### Defaults for HRAM

If you select a customized datalog setup, the default HRAM settings are as follows:

- Use All Domains Setup is checked, and Time Domain: is set to All Domains.
- Trigger On First is set to Cycle.
- Capture is set to All.

These are the defaults whether you use the subset of HRAM controls on the Setup File dialog box or the complete controls available from the Full Edit button.  
If you do not use a customized datalog setup, but instead use one of the standard datalog setups on the Setups tab, the default HRAM settings are as follows:

- Use All Domains Setup is checked, and Time Domain: is set to All Domains.
- Trigger On First is set to Cycle.
- Capture is set to Fail.

The difference is that the default Capture setting for customized setups is All, while for standard setups it is Fail.

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## Summary Report, Lot Definition, and Wafer Information Tabs

### Summary Report, Lot Definition, and Wafer Information Tabs

This section describes the tabs in the DataCollect Setup interface, apart from the Setups tab. The following tabs are described:

- |   |                                       |
|---|---------------------------------------|
| • | <a href="#">Summary Report Tab</a>    |
| • | <a href="#">Lot Definition Tab</a>    |
| • | <a href="#">Wafer Information Tab</a> |

For more information on the Setups tab, see [Setups Tab](#).

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## Summary Report, Lot Definition, and Wafer Information Tabs : Summary Report Tab

### Summary Report Tab

This section describes fields and controls on the Summary Report tab, as well as the Display Options dialog box, which can be called from the Summary Report tab.

Items on the Summary Report tab allow you to:

- Set up how summary data is to be collected and displayed
- Generate a partial or final summary report.

Summary data gives a summary of test execution, including:

- Header information, including lot ID, operator, and test time
- Bin and sort totals, including the number of devices assigned to each bin and sort number, and the percentage of the total assigned to each
- A test summary, including the number of times each test was executed, and the number and percentage of failures
- Detailed information for each test or for each failed test (optional). To display this information, Test Criteria must be All tests or Failed tests. The detailed test information is given in columns with these headings:
  - Number (test number)

- [Name](#) (test name)
- [Total](#) (total number of executions)
- [Failures](#) (number of failures)
- [%Part](#) (percentage of failed parts)
- [%Exec](#) (percentage of failed executions)

The following topics are included:

- [Output Controls](#)
- [Display Controls](#)
- [Report Controls](#)
- [Final Summary/End Lot and Validation](#)

The Summary Report tab looks like this:

**DataCollect Setup - Standard Setups**

Setups | **Summary Report** | Lot Definition | Wafer Information

**Output**

☒ Window

☒ Text File

☒ STDF File

Test Criteria:

Output Width:

☐ Report By Test Name

**Report**

☒ Total Summary (By Test)

☐ Total Summary (By Site)

## Summary Report Tab

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Summary Report, Lot Definition, and Wafer Information Tabs : Summary Report Tab : Output Controls

## Output Controls

The Output frame of the Summary Report tab looks like this:

Output

☒ Window

☒ Text File

MP4.xls\_sum\_03282012\_081304.txt

Browse...

☒ STDF File

MP4.xls\_sum\_03282012\_081305.std

Browse...

Test Criteria:

All tests

Output Width:

0

☐ Report By Test Name

## Summary Report Tab Output Frame

This frame includes controls for:

- [Output Destination](#)
- [Test Criteria](#)
- [Output Width](#)
- [Report By Test Name](#)

## Output Destination

These controls specify the summary report output. You can choose one or more of the following options:

|           |                                        |
|-----------|----------------------------------------|
| Window    | Write to the OutputWin window.         |
| Text File | Write to an ASCII file.                |
| STDF File | Write to a binary file in STDF format. |

Select STDF or Text File to enable the field and use the Browse button to specify a filename. If you do not specify a filename, a default is used.

If the same STDF file is specified for a datalog stream output and the summary output, an error is returned to the output window. The specified name is used for the summary STDF file, and a new name is generated for the datalog stream STDF file.

For more information, see Output Options.

Test Criteria

This drop-down specifies which tests are to be included in the summary report:

|              |                                                |
|--------------|------------------------------------------------|
| All tests    | Include detailed information for all tests.    |
| Failed tests | Include detailed information for failed tests. |
| No tests     | Do not include test reports in the summary.    |

Output Width

This control specifies the maximum number of characters to be displayed per line. The default is 0, which means "no wrapping."

Report By Test Name

This control specifies whether the summary reports is to use the test name. By default, the summary uses the Flow name.

You can specify the test name in:

- The TName column of the Flow Table sheet
- The TName parameter of TheExec.Flow.TestLimit

By default, if this box is not checked, the test name appearing in the summary is taken from the Flow Table. If this box is checked, the test name is taken from TestLimit.

Reporting by test name can help in reconciling the summary report to tests.

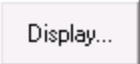
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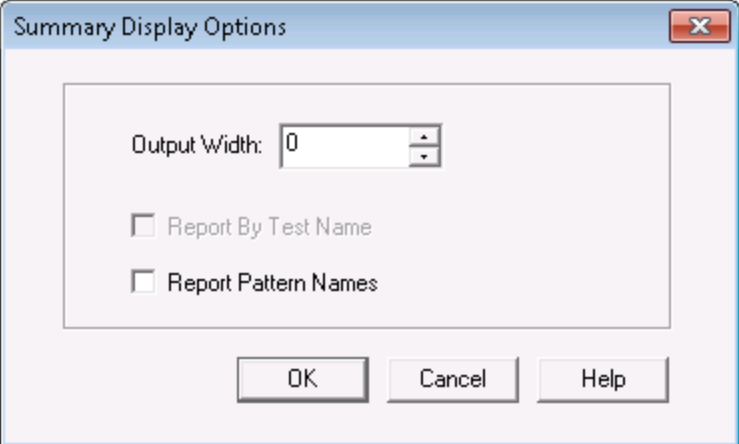
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Summary Report, Lot Definition, and Wafer Information Tabs : Summary Report Tab : Display Controls

### Display Controls

 Use this button to bring up a dialog box for specifying display options:



The dialog box titled "Summary Display Options" contains the following controls:

- Output Width:** A text box with the value "0" and a spinner control.
- Report By Test Name:** An unchecked checkbox.
- Report Pattern Names:** An unchecked checkbox.
- Buttons:** OK, Cancel, and Help.

### Summary Display Options Window

The options are as follows:

#### Output Width

This control specifies the maximum number of characters to be displayed per line. The default is 0, which means "no wrapping."

#### Report Pattern Names

This check box specifies whether the pattern name is to be included in the summary names of functional tests.

By default, the summary output for a functional test displays only the test name as specified in the Flow Table. This can make it difficult to determine what was actually tested in a given test, especially when a single functional test executed many patterns. If this option is checked, the name of the pattern or pattern set that was tested is included as part of the test name.

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Summary Report, Lot Definition, and Wafer Information Tabs : Summary Report Tab : Report Controls

## Report Controls

The Report frame of the Summary Report tab let you generate a partial or final summary report, including summaries only by test or by test and by site:

Report

☒ Total Summary (By Test)

☐ Total Summary (By Site)

Get Partial Summary

Get Final Summary/End Lot

## Summary Report Frame

The options are as follows:

Total Summary (By Test) and (By Site)

The Total Summary buttons specify the format of summary reports:

**(By Test)**

The summary report includes a summary of test execution with this information:

- Header information, including lot ID, operator, and test time
- A test summary listed by test, including the number of times each test was executed, and the number and percentage of failures.
- The fields of information given for each test include: bin and sort totals, including the number of devices assigned to each bin and sort number, and the percentage of the total assigned to each

**(By Site)**

The summary report includes all the information listed for (By Test), plus site-specific summaries. Summaries for each site are grouped together.

**Get Partial Summary**

Click this button to generate an interim summary report at any time when test execution is paused between devices. The summary report is cumulative, summarizing the results since the beginning of the test and including the last report summary.

As soon as you click this button:

- The report is generated immediately, not after you click OK.

- The DataCollect Setup window closes.

**Get Final Summary/End Lot**

Click this button to indicate that the lot is finished testing. It generates a final summary report and sends it to the selected output destinations. If you have selected Text File or STDF File as the output, the final summary is written to the file and the file is closed. This button clears the previous summary data. On the next invocation, a summary report will contain only the recently gathered summary data, if any.

As soon as you click this button:

- The report is generated immediately, not after you click OK.

- The DataCollect Setup window closes.

For more information, see [Final Summary/End Lot and Validation](#).

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Summary Report, Lot Definition, and Wafer Information Tabs : Summary Report Tab : Final Summary/End Lot and Validation

Final Summary/End Lot and Validation

The Summary Report tab has the button Get Final Summary/End Lot. Both the final summary and end lot have VBT syntax equivalents.

Effects of End Lot

Request end lot on the Summary Report tab by clicking the Get Final Summary/End Lot button. End Lot resets the device number to 1. In addition, the End Lot method clears (that is, sets to 0) the following information:

- |   |                                                |
|---|------------------------------------------------|
| • | Failed tests count and failed parts count      |
| • | Passed tests count and passed parts count      |
| • | Executed tests count and executed parts count  |
| • | Alarmed tests count and alarmed parts count    |
| • | Retested tests counts and retested parts count |
| • | Sort count                                     |
| • | Bin count                                      |

- Data counts related to wafer sort:
  - Tested parts count
  - Passed parts count
  - Retested parts count

End Lot closes the STDF and text output files properly. End Lot also deletes the filenames, either user-specified or default, so that the next run can use new filenames. On coordinating postprocessing with End Lot, see [Postprocessing Datalog Output Files](#) in the section on datalog output.

The Wafer Results Record (WRR), which contains data counts related to wafer tests, is generated on the EndOfWafer event. For more information, see [Wafer Results Record \(WRR\)](#).

End Lot and the DataCollection Production Control

IG-XL Production Controls include the DataCollection control, which lets the DataCollect Setup GUI be called from a user-designed operator interface. It also permits programmatic access to the DataCollect Setup settings. When using this control, clicking [Get Final Summary/End Lot](#) on the Summary Report tab (or setting the LotSetup.EndLot property) is not the same as executing the EndLot method of the DataCollection control.

For more information, see [DataCollection Control](#) and [EndLot Method](#) in the Production Controls section.

Revalidation

The effects of end lot and revalidation are similar but not necessarily identical. The Setups tab has a control for specifying the action to be taken when revalidation occurs during a test program run. These actions also affect the files that are output for the summary report. For more information, see [Following Validate/Save](#).

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Summary Report, Lot Definition, and Wafer Information Tabs : Lot Definition Tab

## Lot Definition Tab

Items on the Lot Definition tab let you:

- Specify information for the lot being tested, such as the header of output reports and data for the STDF file
- Specify a sample rate and options relating to disk space, error messages, and the buffer.

This section describes fields and controls on the Lot Definition tab, as well as the More Information and Sampling Setup dialog boxes, which can be called from the Lot Definition tab. The following topics are included:

- Lot Info Frame
- Device Number Control
- Comment Control
- Lot Options Frame
- More Information Dialog Box

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Summary Report, Lot Definition, and Wafer Information Tabs : Lot Definition Tab : Lot Info Frame

## Lot Info Frame

This frame provides controls for entering information for the header of output reports and for the Master Information Record (MIR) of the output STDF file:

Lot Info

Lot ID: 1813

Test Mode: Engineering

Operator: JoeGreen

Part Type: FrzDIDstn

More ...

## Lot Info Frame

For information on STDF fields that are filled in from this tab, see Master Information Record (MIR) in the STDF Specification.

Some lot information is provided automatically: for example, tester name and test date and time. For information on STDF fields that are filled in by DataCollect, see Master Information Record (MIR) in STDF Support for IG-XL.

To provide the MIR information, fill in these fields with any text string, or select from the list:

|           |                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lot ID    | Enter the ID.                                                                                                                                                                      |
| Test Mode | The drop-down list contains the values defined in the STDF specification for the Mir.MODE_CODE field, including the characters 0-9, which can be used for customer-defined values. |
| Operator  | If you leave this blank, your username is used.                                                                                                                                    |
| Part Type | Enter the part type.                                                                                                                                                               |

You can leave fields blank; only Operator has a default. The values you enter are not checked for validity.

You can also enter additional information for the lot to be stored in the MIR or the Site Description Record (SDR). Click More to bring up a dialog box for entering these values. See More Information Dialog Box.

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Summary Report, Lot Definition, and Wafer Information Tabs : Lot Definition Tab : Device Number Control

## Device Number Control

The Device Number control specifies the device ID for the next device to be tested. This value is written to the Part Results Record (PRR) of the STDF file.

|                |                 |
|----------------|-----------------|
| Device Number: | <div>1901</div> |
|----------------|-----------------|

## Device Number Control

The device number is automatically incremented by 1 each time a device is tested. You can enter a nondefault device number in this field. The number you enter is automatically incremented by 1 as each device is tested. For more information, see [Part Results Record \(PRR\)](#) in the STDF Support for IG-XL.

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Summary Report, Lot Definition, and Wafer Information Tabs : [Lot Definition Tab](#) : Comment Control

## Comment Control

This text box specifies a comment to be displayed in the output window:

|          |                              |
|----------|------------------------------|
| Comment: | <div>Check for Simon B</div> |
|----------|------------------------------|

## Comment Control

This is valid only if the Setups tab has selected the Window output destination. The maximum comment length is 80 characters.

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Summary Report, Lot Definition, and Wafer Information Tabs : Lot Definition Tab : Lot Options Frame

### Lot Options Frame

This frame contains setup options concerning disk space, error messages, and the buffer:

Options

Show Message When Disk Space Is Lower Than Minimum

☒ After Setting Up Data Collection

☒ During Execution

Minimum Available Disk Space  MB

☒ Show Message When Write Errors Occur

☒ Flush Buffer During Execution Every  record(s)

### Options Frame

The options are listed here and described in the following sections:

- Show Message When Disk Space Is Lower Than Minimum
- Show Message When Write Errors Occur
- Flush Buffer During Execution

### Show Message When Disk Space Is Lower Than Minimum

Datalogging requires a certain amount of disk space for datalog files. If the system runs out of disk space, an error is returned to the output window, the STDF and text files are closed, and the flow stops execution. If datalog data cannot be written to a file, it may be lost.

You can use this option to have a warning displayed if the available disk space is less than a minimum you specify. To enable the option, check one or both of the boxes under the option label and specify a minimum using the Minimum Available Disk Space control. The check boxes cause the disk space to be checked at different times:

|                                  |                                                                                                                                                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| After Setting Up Data Collection | Checks the available disk space after you have set up data collection but before the test is run. If the disk space is lower than the minimum, a warning appears.    |
| During Execution                 | Checks the available disk space during test execution. (It checks only for writing datalog data, not summary data.) See the description in the following paragraphs. |

If you select **During Execution**, and if a check reveals that the disk space is lower than the minimum, a message appears stating this. It also asks "Do you want to show this message in the future?" You have these options:

|     |                                                                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No  | The datalog system stops running the disk space check. If you finally run out of disk space, an error is returned and the flow stops.                                                         |
| Yes | The datalog system continues to check for disk space. It brings up a warning if the check, using an algorithm, determines that there is not enough space to datalog the devices being tested. |

In production, you typically set the warning limit higher to provide enough time and opportunity to fix the problem. If you select **Yes**, a warning occurs if there is not enough disk space. To resume testing after the warning message, you can clear enough disk space so that the Minimum Available Disk Space you have specified is met. You can then issue an **ApplySetup** or click OK in the GUI. At this point, you can continue testing.

#### Show Message When Write Errors Occur

If a write error occurs (for example, if the disk is full or a network directory is not available), datalogged data cannot be written and may be lost. Enabling this option causes a warning to be displayed if a write error occurs.

If a write error occurs, a warning appears asking if you want to stop execution of the test. If you answer **Yes**, execution stops after the current device is binned; execution is not halted immediately.

#### Flush Buffer During Execution

Flushing the write buffer to disk can minimize the amount of data that would be lost in case of a write error. To enable the option, check the box and enter the frequency at which the buffer should

be flushed. The default is every 100 records.

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Summary Report, Lot Definition, and Wafer Information Tabs : [Lot Definition Tab](#) : [More Information](#)  
Dialog Box

## More Information Dialog Box

This dialog box lets you enter additional information about a lot. Values are written to the STDF file and optionally to the OutPutWin window and the text file.

Invoke this dialog box by clicking [More](#) on the Lot Definition tab. The dialog box displays two tabs:

|     |                           |
|-----|---------------------------|
| MIR | Master Information Record |
| SDR | Site Description Record.  |

For values that will be written to the STDF file, enter the value in the appropriate field. If a field has a limited set of valid values, the values are available on a drop-down list.  
For more information, see:

- [Lot Definition Tab](#)
- [Master Information Record \(MIR\)](#)
- [Site Description Record \(SDR\)](#)

## Writing Values to the Window and Text File

By default, values that you enter here are written only to the STDF file. They do not appear in the OutPutWin window or in the ASCII text file.

To have a value written to the window and text file as well as to the STDF file, click the box next to the field.

For example, the following writes the PKG Type to the STDF file, window, and ASCII text file, but the Process ID only to the STDF file:

More information

MIR | SDR

Master Information

|              |  |                          |              |       |                                     |
|--------------|--|--------------------------|--------------|-------|-------------------------------------|
| Aux File:    |  | <input type="checkbox"/> | PKG Type:    | LTvt  | <input checked="" type="checkbox"/> |
| Burn Time:   |  | <input type="checkbox"/> | Process ID:  | Rgltt | <input type="checkbox"/>            |
| Command:     |  | <input type="checkbox"/> | Prot Code:   |       | <input type="checkbox"/>            |
| Date Code:   |  | <input type="checkbox"/> | Retest Code: |       | <input type="checkbox"/>            |
| Design Rev:  |  | <input type="checkbox"/> | ROM ID:      |       | <input type="checkbox"/>            |
| Eng ID:      |  | <input type="checkbox"/> | Serial Num:  |       | <input type="checkbox"/>            |
| Exec Type:   |  | <input type="checkbox"/> | Setup ID:    |       | <input type="checkbox"/>            |
| Exec Ver:    |  | <input type="checkbox"/> | Spec Name:   |       | <input type="checkbox"/>            |
| Facility ID: |  | <input type="checkbox"/> | Spec Ver:    |       | <input type="checkbox"/>            |
| Family ID:   |  | <input type="checkbox"/> | Sub Lot:     |       | <input type="checkbox"/>            |
| Floor ID:    |  | <input type="checkbox"/> | Supervisor:  |       | <input type="checkbox"/>            |
| Flow ID:     |  | <input type="checkbox"/> | Test Code:   |       | <input type="checkbox"/>            |
| Job Rev:     |  | <input type="checkbox"/> | Test Temp:   |       | <input type="checkbox"/>            |
| Oper Frq:    |  | <input type="checkbox"/> | User Text:   |       | <input type="checkbox"/>            |

Enter a value to write to STDF file.  
Check box to write to window and text file as well.

OKCancelHelp

Lot More information Window

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## dltabs.4.13

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|                                                                                   |                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|

[Summary Report, Lot Definition, and Wafer Information Tabs](#) : Wafer Information Tab

---

### Wafer Information Tab

Use this tab to enter wafer configuration data:

**DataCollect Setup - Standard Setups**

Setups | Summary Report | Lot Definition | **Wafer Information**

Wafer ID:  Diameter Of Wafer:

Height Of Die:  Width of Die:

Unit:  Orientation Of Wafer Flat:

**Coordinate of Center Die**

X:  Y:

**Direction of Wafer**

X:  Y:

OK Cancel Help

### Wafer Information Tab

Most of this data is stored in the STDF file in a Wafer Configuration Record (WCR). The Wafer ID is used in the Wafer Information Records (WRRs) and Wafer Results Records (WRRs).

Use of this tab is meaningful only if you specify the STDF file as an output option.

Fields on the Wafer Information tab are as follows:

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| Wafer ID          | A string to identify the wafer being tested.                     |
| Diameter of Wafer | Diameter of the wafer, in the units specified in the Unit field. |

|                                |                                                                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Height of Die,<br>Width of Die | Height and width of each die, in the units specified in the Unit field.                                                                                                                                                                  |
| Unit                           | <p>Unit to be used with the number specified in the Diameter, Height, and Width fields. The options are:</p> <ul style="list-style-type: none"><li>Unknown</li><li>Inches</li><li>Centimeters</li><li>Millimeters</li><li>Mils</li></ul> |
| Orientation of Wafer<br>Flat   | <p>The options are:</p> <ul style="list-style-type: none"><li>Unknown</li><li>Up</li><li>Down</li><li>Left</li><li>Right</li></ul>                                                                                                       |
| Coordinate of Center<br>Die    | The X and Y coordinates. Enter a number, or use the up and down arrows to display the number.                                                                                                                                            |
| Direction of Wafer             | <p>Positive X and Y directions of the wafer. X options are:</p> <ul style="list-style-type: none"><li>Unknown</li><li>Right</li></ul>                                                                                                    |

- [Left](#)

Y options are:

- [Unknown](#)
- [Up](#)
- [Down](#)

For more information, see:

- |                                                    |
|----------------------------------------------------|
| • <a href="#">Output Options</a>                   |
| • <a href="#">Wafer Configuration Record (WCR)</a> |
| • <a href="#">Wafer Information Record (WIR)</a>   |
| • <a href="#">Wafer Results Record (WRR)</a>       |